

Conformal Prediction for Hierarchical Time Series with the Use Case of EV charging

Guillaume Principato

Joint work with Yvenn Amara-Ouali, Yannig Goude, Bachir Hamrouche, Jean-Michel Poggi and Gilles Stoltz.





EVS37

COEX, Korea, April 23-26, 2024

Quantifying the uncertainty of EV charging with probabilistic load forecasting

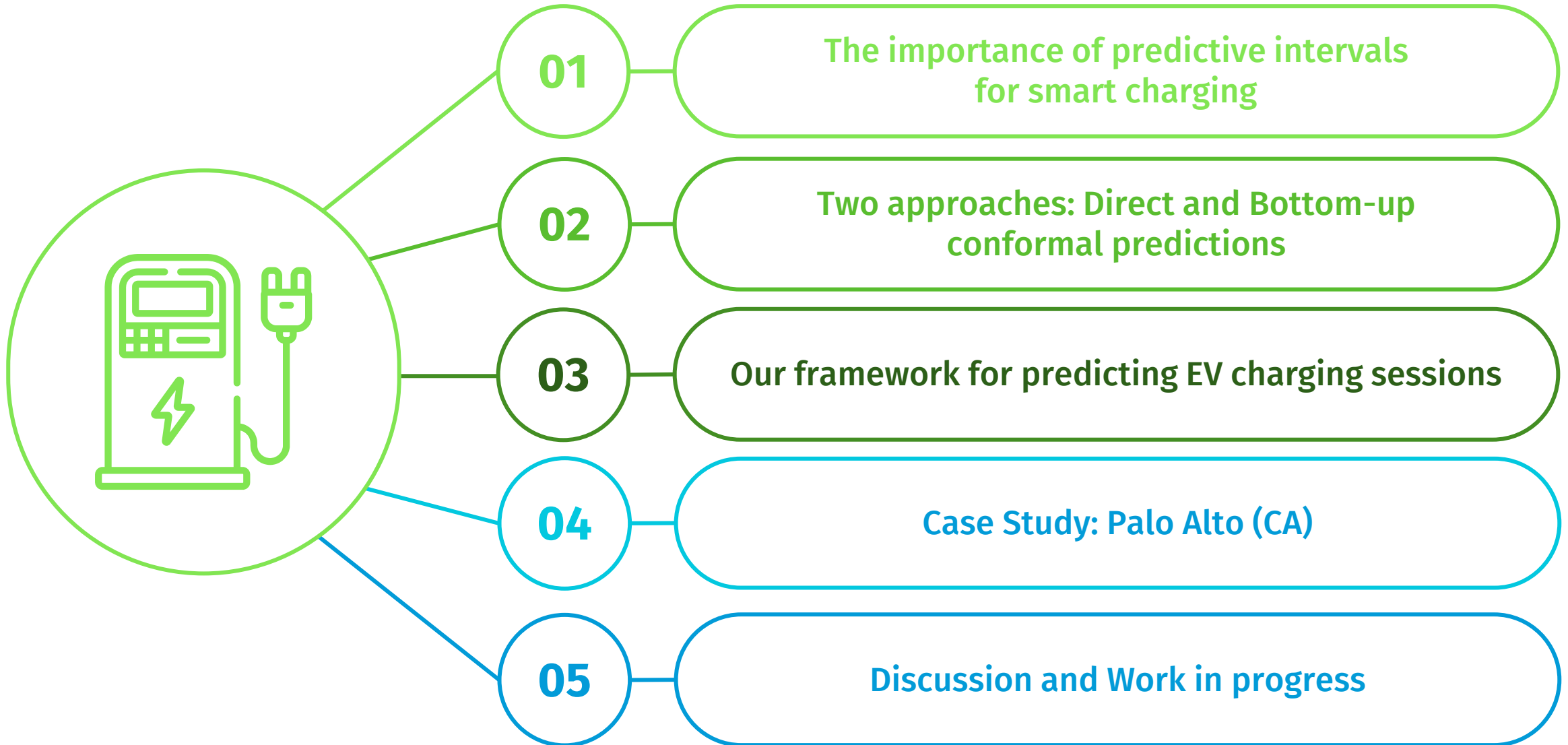
Yvenn Amara-Ouali^{1,2}, Bachir Hamrouche¹, Guillaume Principato^{1,2,3}, Yannig Goude^{1,2}

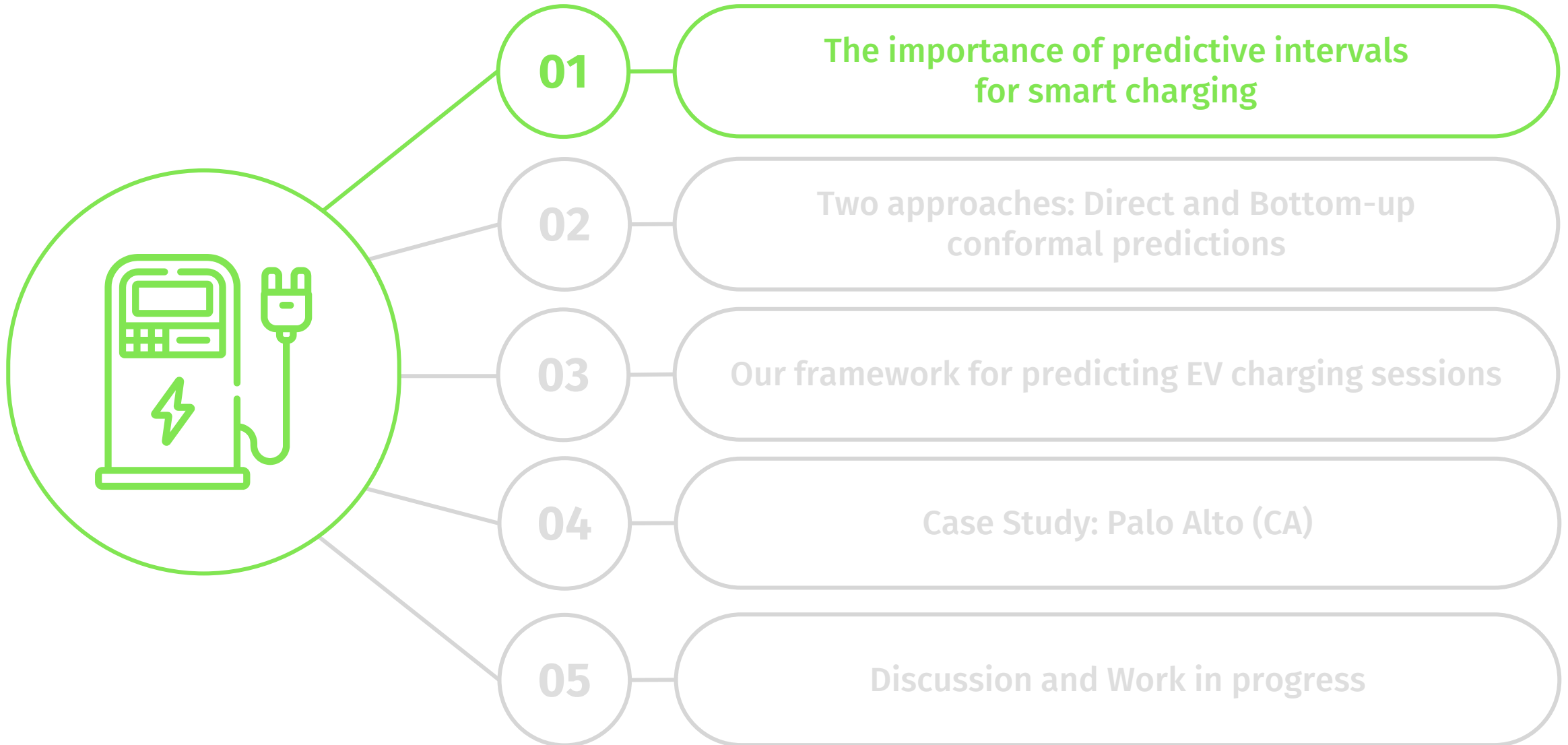
¹EDF R&D, Palaiseau, FRANCE

²University Paris-Saclay, LMO, Orsay, FRANCE

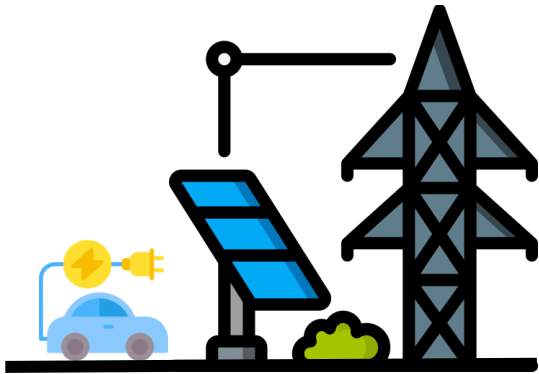
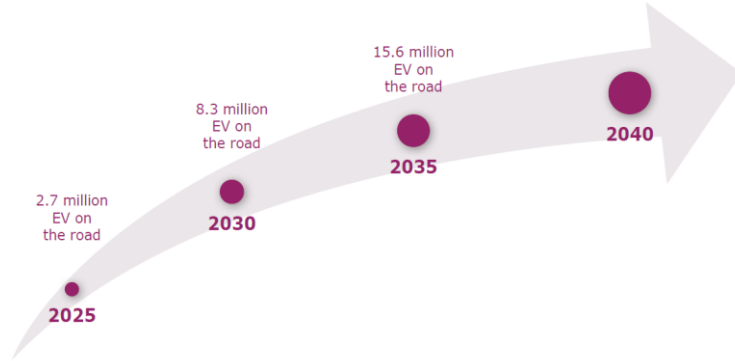
³Inria, CELESTE, Orsay, FRANCE

yvenn.amara-ouali@edf.fr





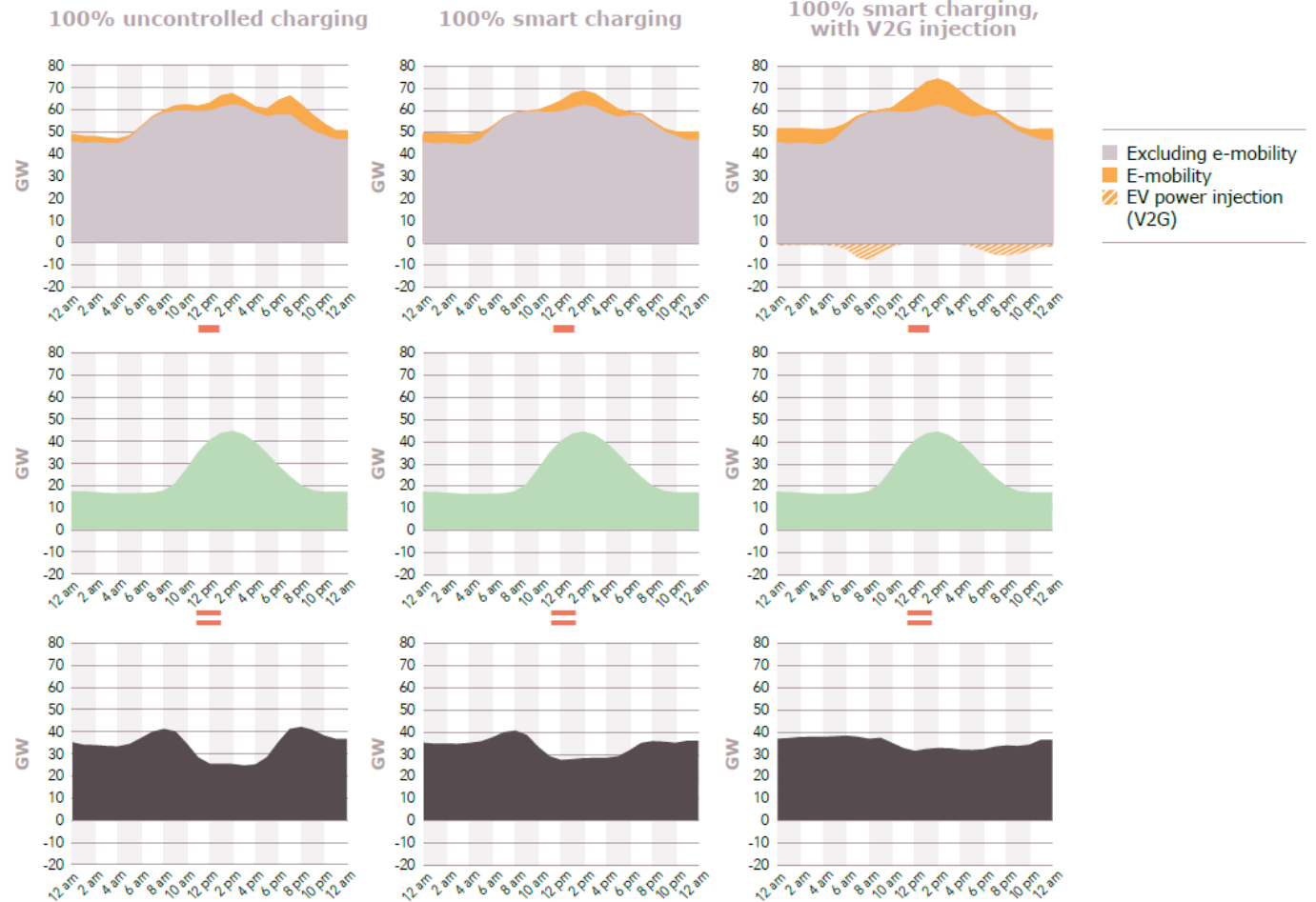
The importance of understanding EV charging behaviours



Total demand

Non-dispatchable renewable power production

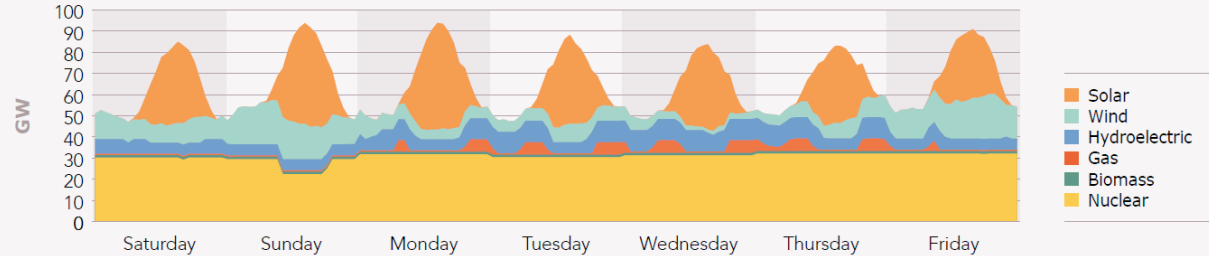
Residual (net) demand met with fossil fuel, hydro and nuclear generation or imports



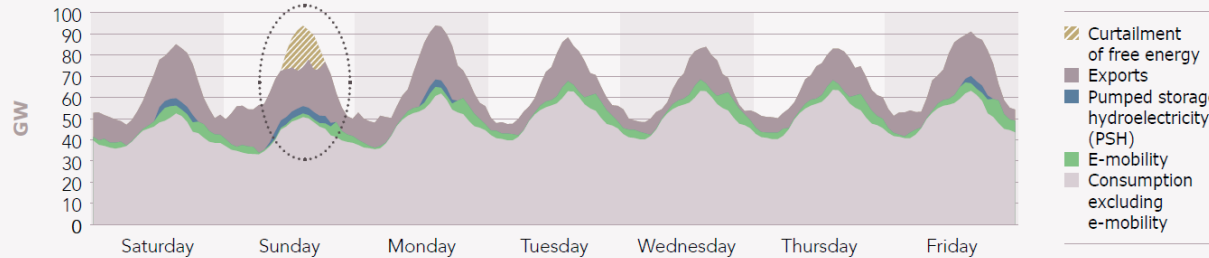
The importance of understanding EV charging behaviours

100% UNCONTROLLED CHARGING

Production in France during one week in July

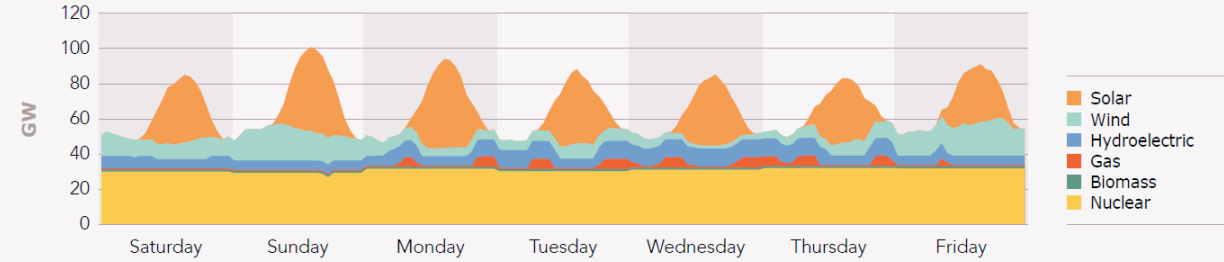


Demand in France during one week in July

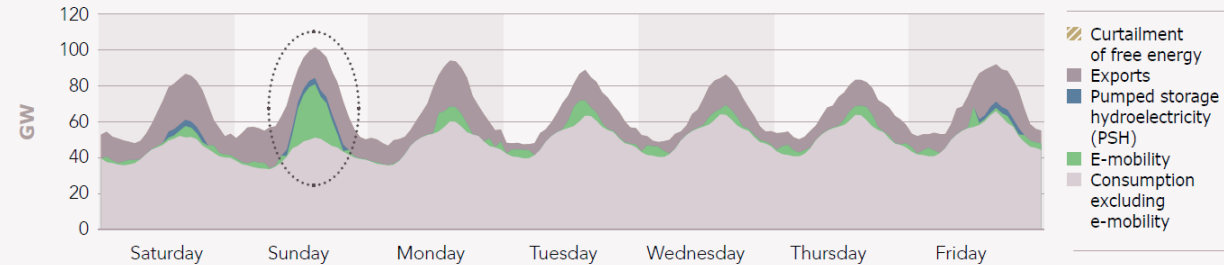


100% SMART CHARGING

Production in France during one week in July



Demand in France during one week in July



Problematic

Producing **accurate forecasts** of EV charging demand
to enable **resilience** for the grid

EV transactions at charging points

Sessions

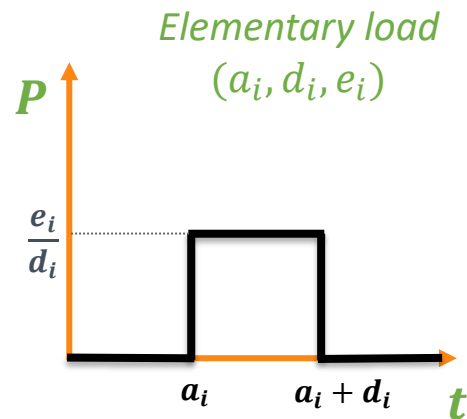
- The data often comes as a history of charging sessions
- Charging session = charging event for one vehicle

$$\text{Session } i = (a_i, d_i, p_i, e_i)$$

- a_i : arrival time
- d_i : charge duration
- p_i : parking duration
- e_i : energy demand

Elementary load profiles

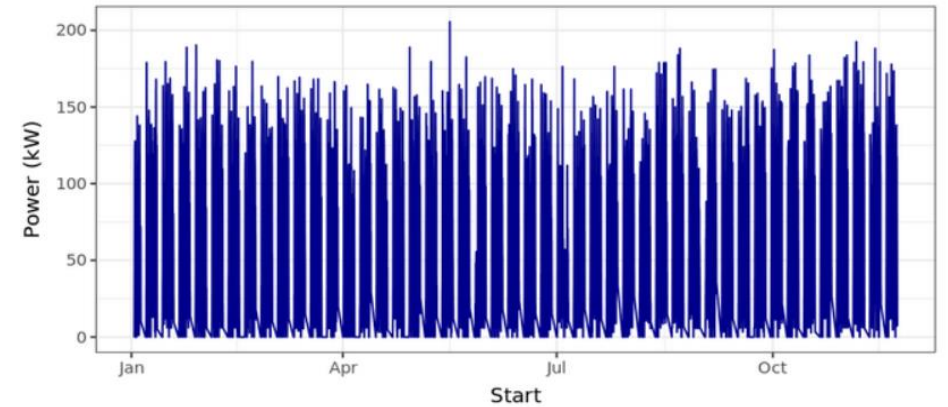
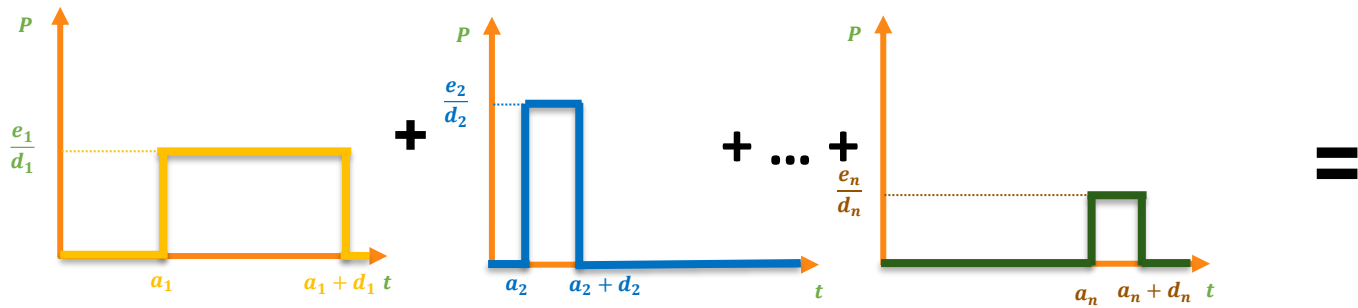
- For each session, an elementary load curve can be reconstructed from the session variables



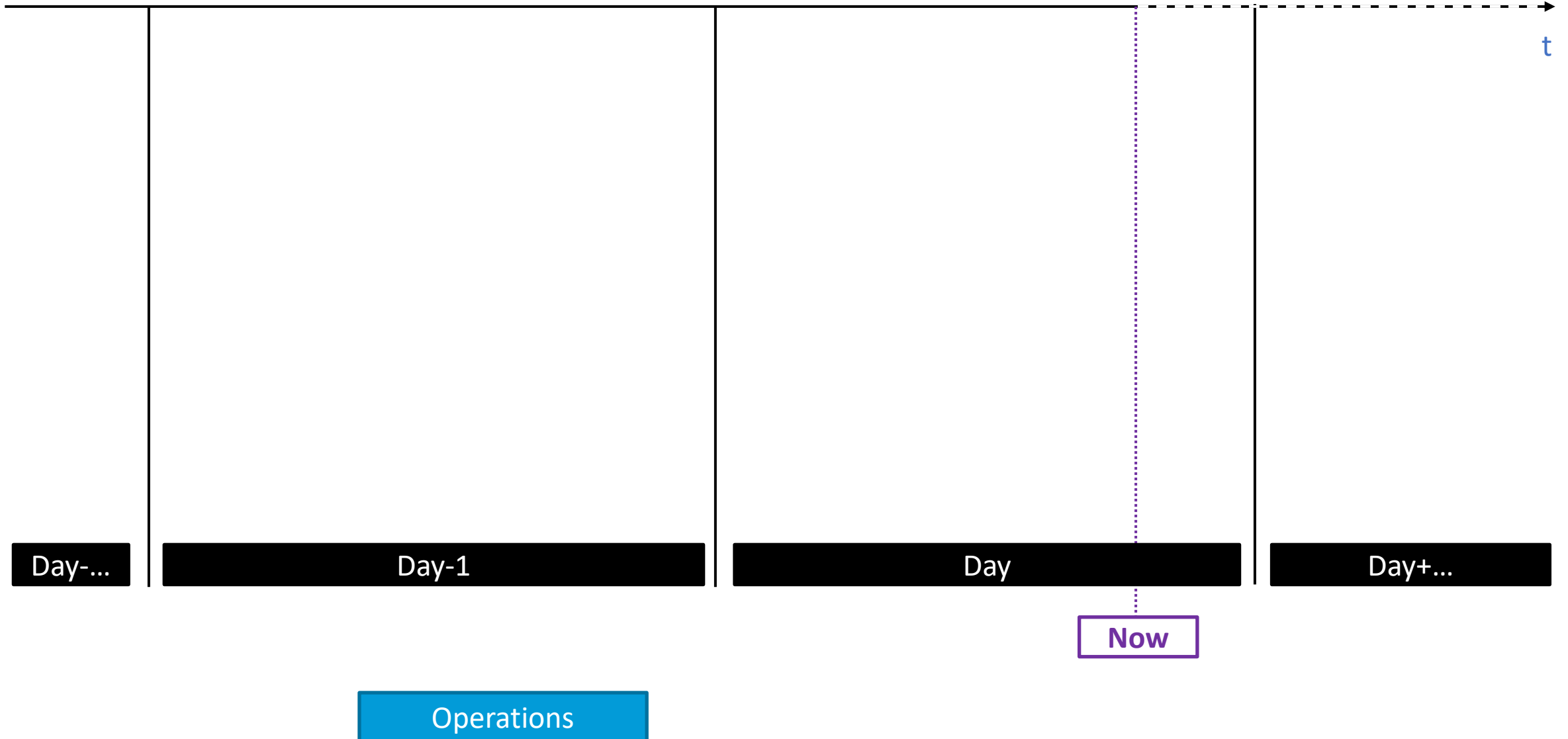
Profile reconstruction

Global load

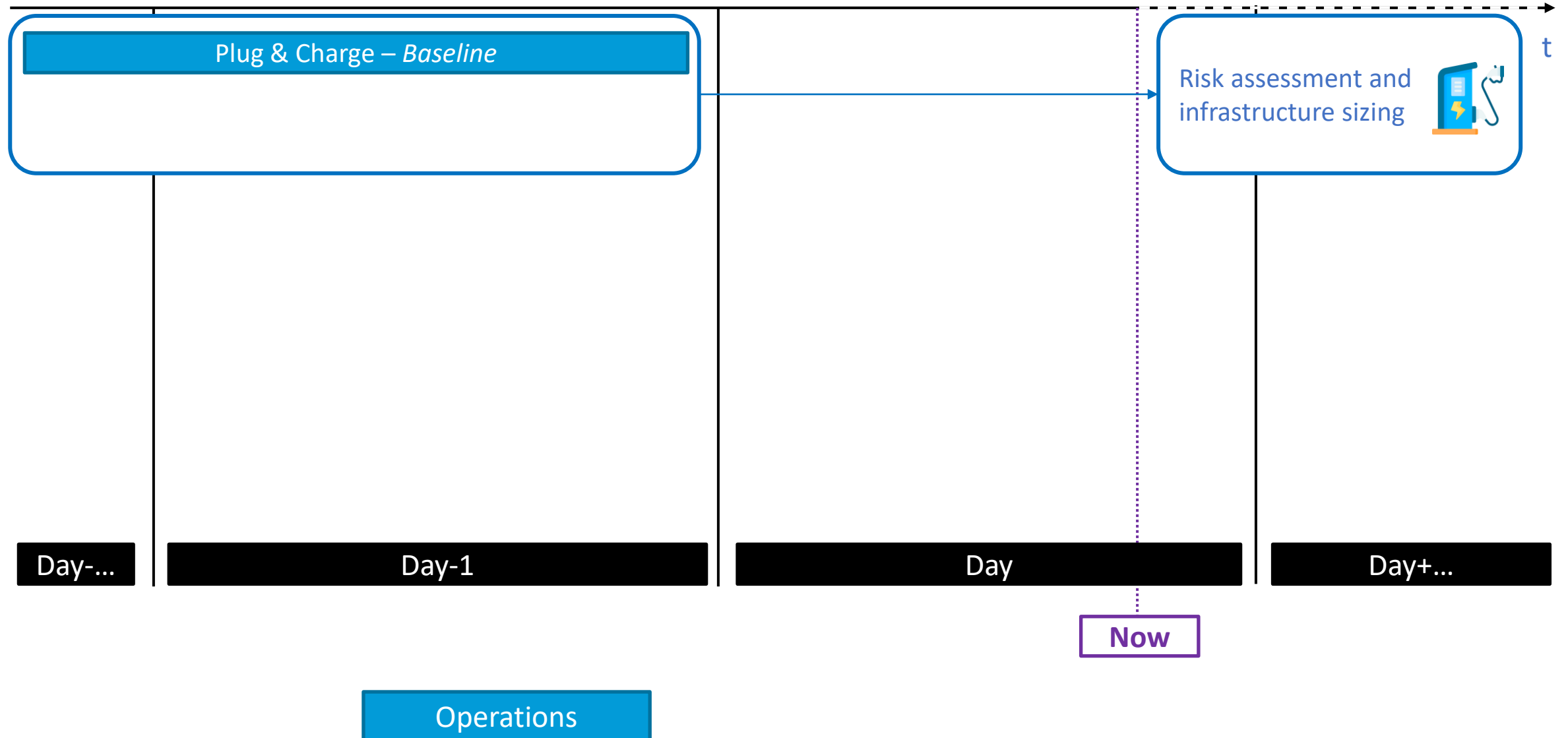
- Adding the elementary profiles leads to global profiles of interest at the scope of a station or dataset



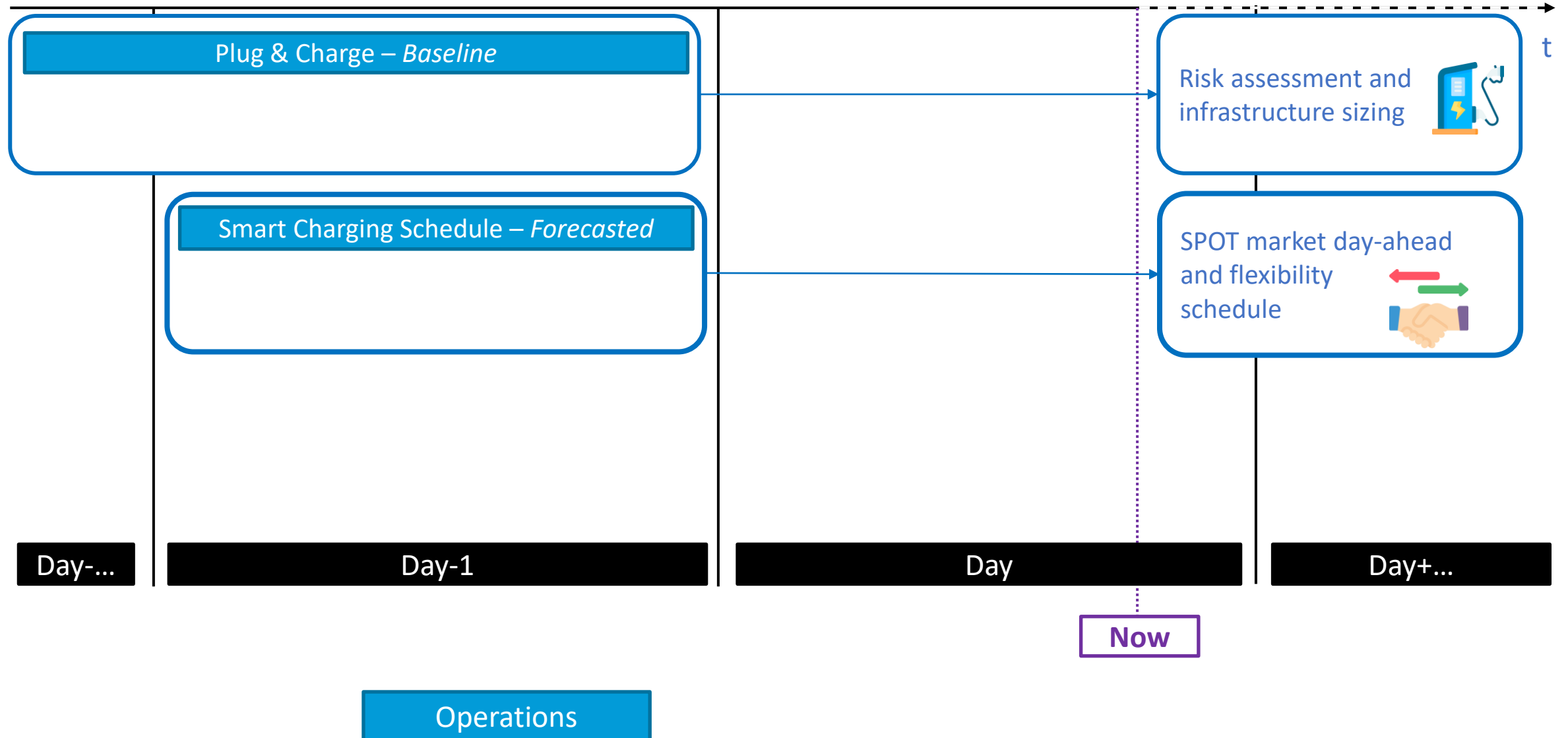
Operations



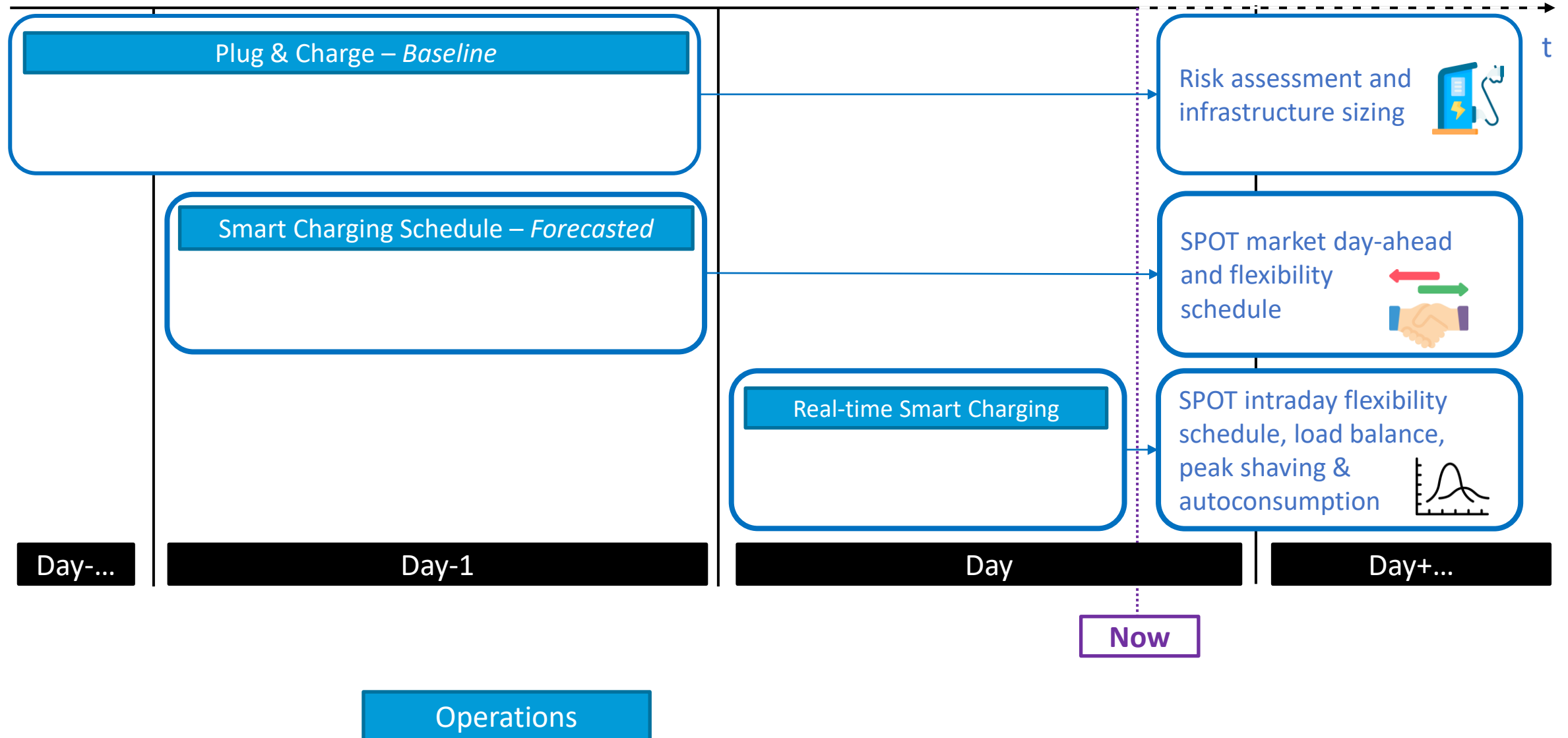
Operations



Operations



Operations



Building up on previous work

Main Results

- Automatic framework for assessing the performances of a wide variety of models across all datasets
- Direct approaches perform better than bottom-up approaches overall
- Bottom-up approaches outperform direct approaches on residential data

Limitations

- Exogenous information such as traffic and weather not considered

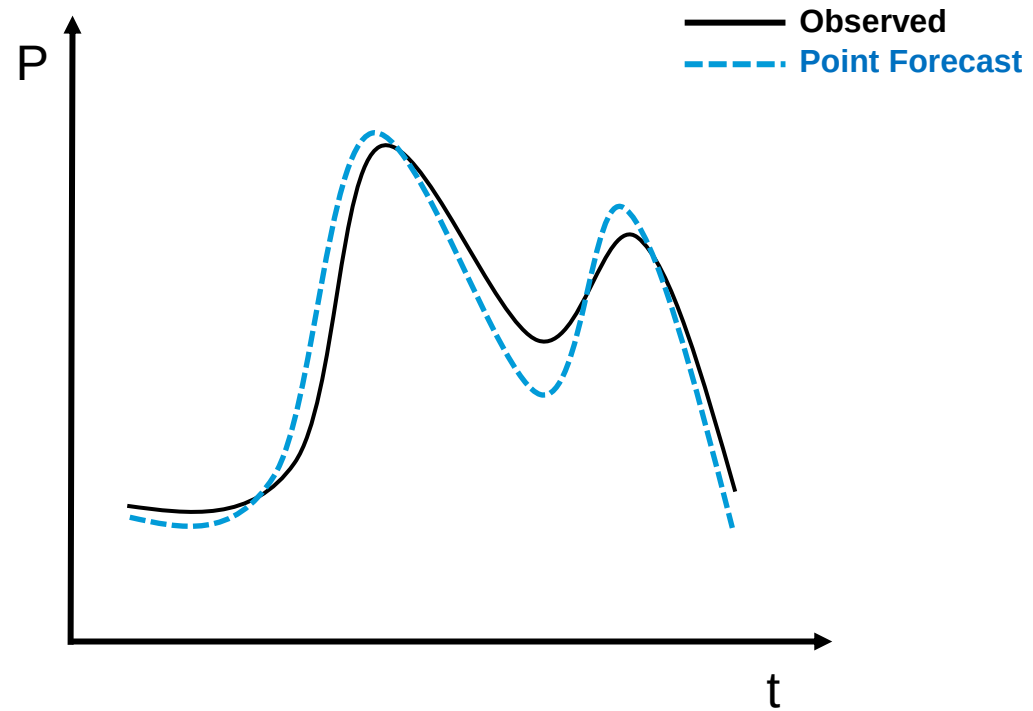
Perspectives

- Using the hierarchical structure of the data to improve bottom-up forecasts
- **Probabilistic forecasts of the EV charging demand**
- Improving the modelling performance on different datasets altogether (e.g., transfer learning)

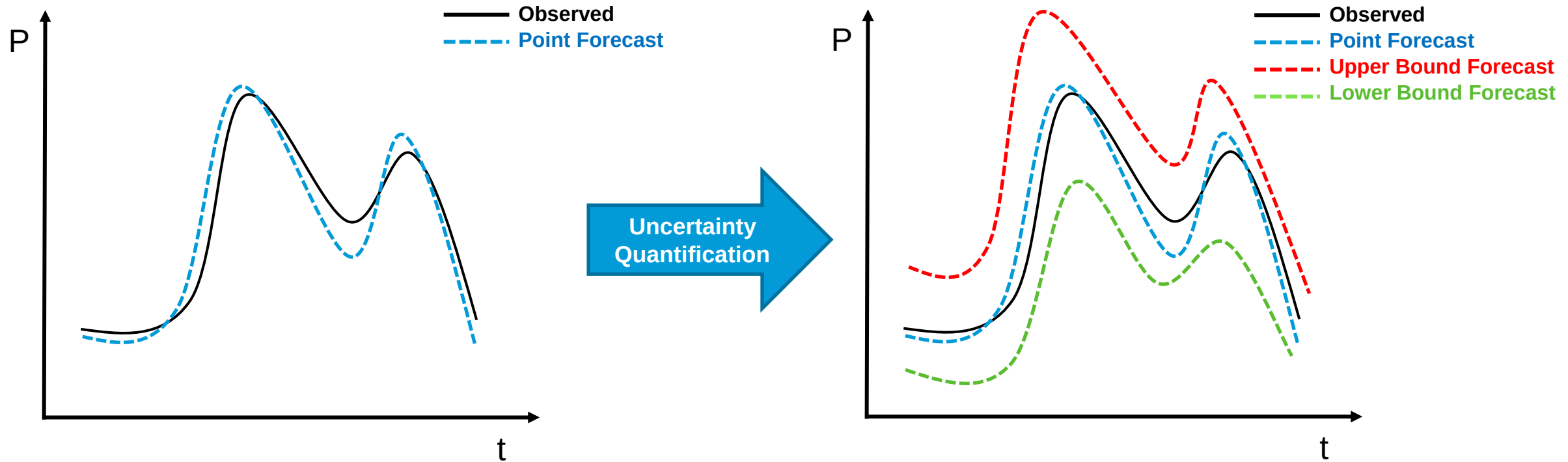
Publication

Amara-Ouali, Y., Goude, Y., Hamrouche, B., & Bishara, M. (2022). A benchmark of electric vehicle load and occupancy models for day-ahead forecasting on open charging session data. In *Proceedings of the Thirteenth ACM International Conference on Future Energy Systems* (pp. 193-207).

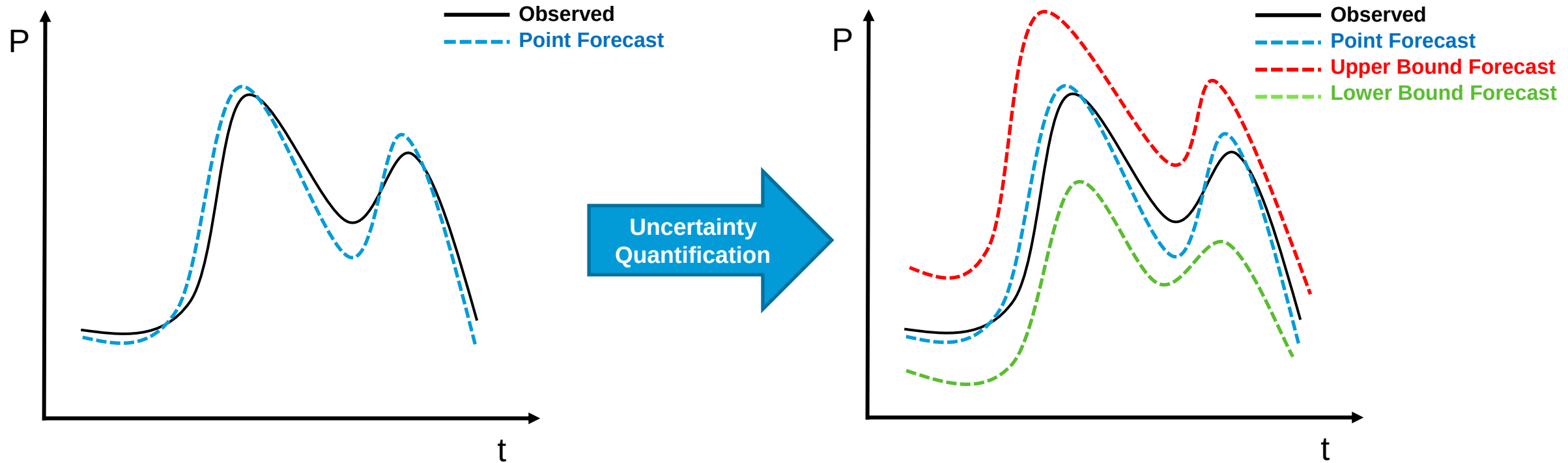
Predicting intervals



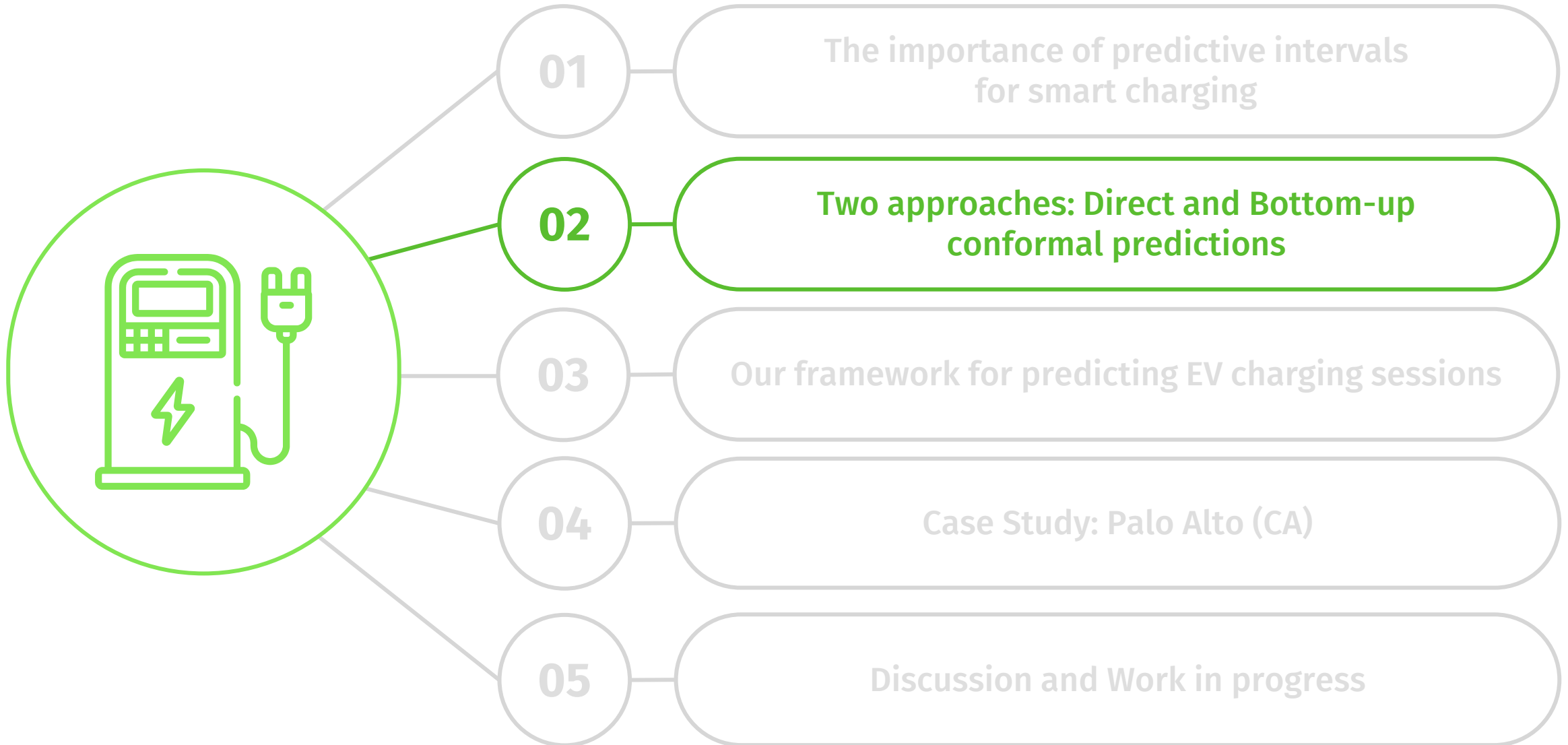
Predicting intervals



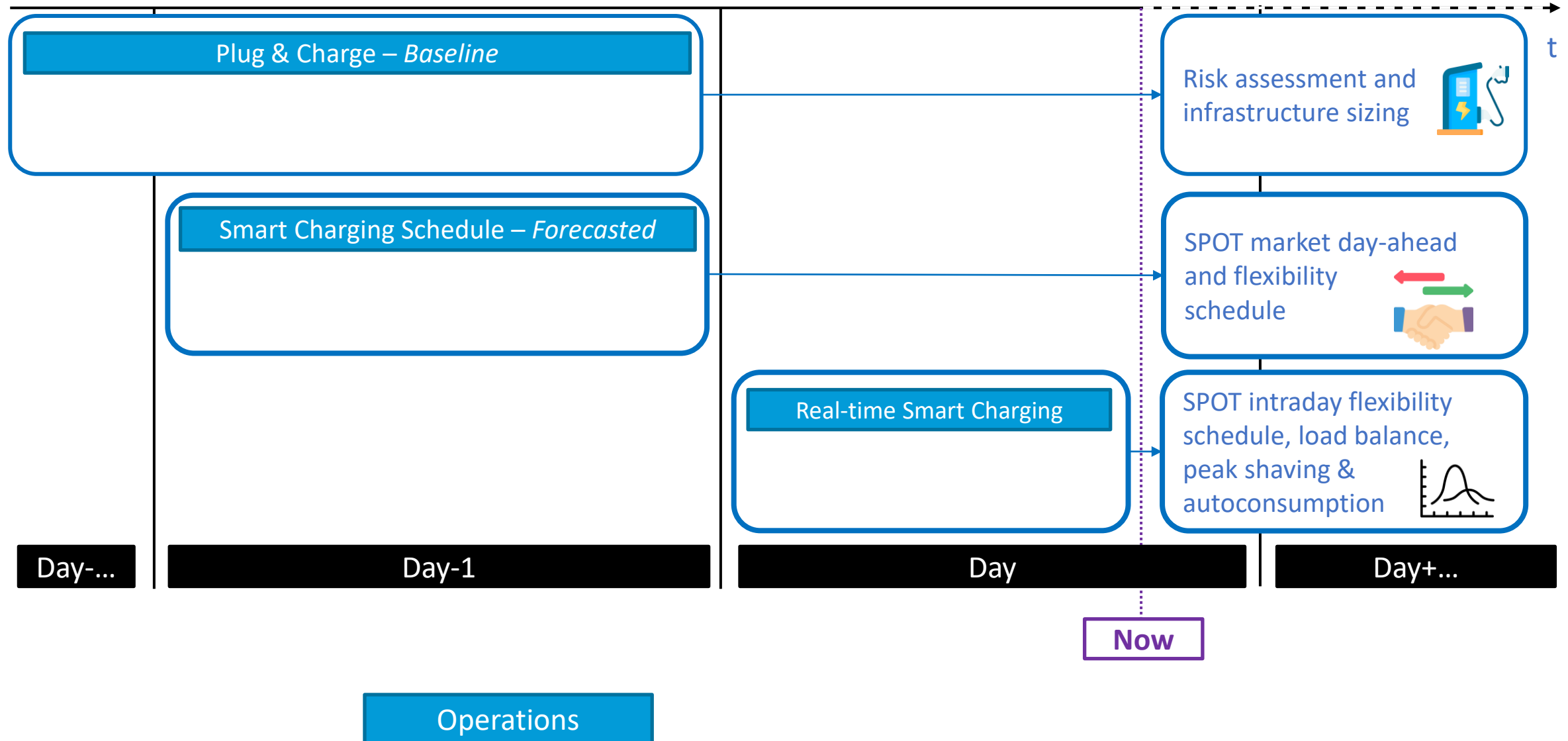
Predicting intervals



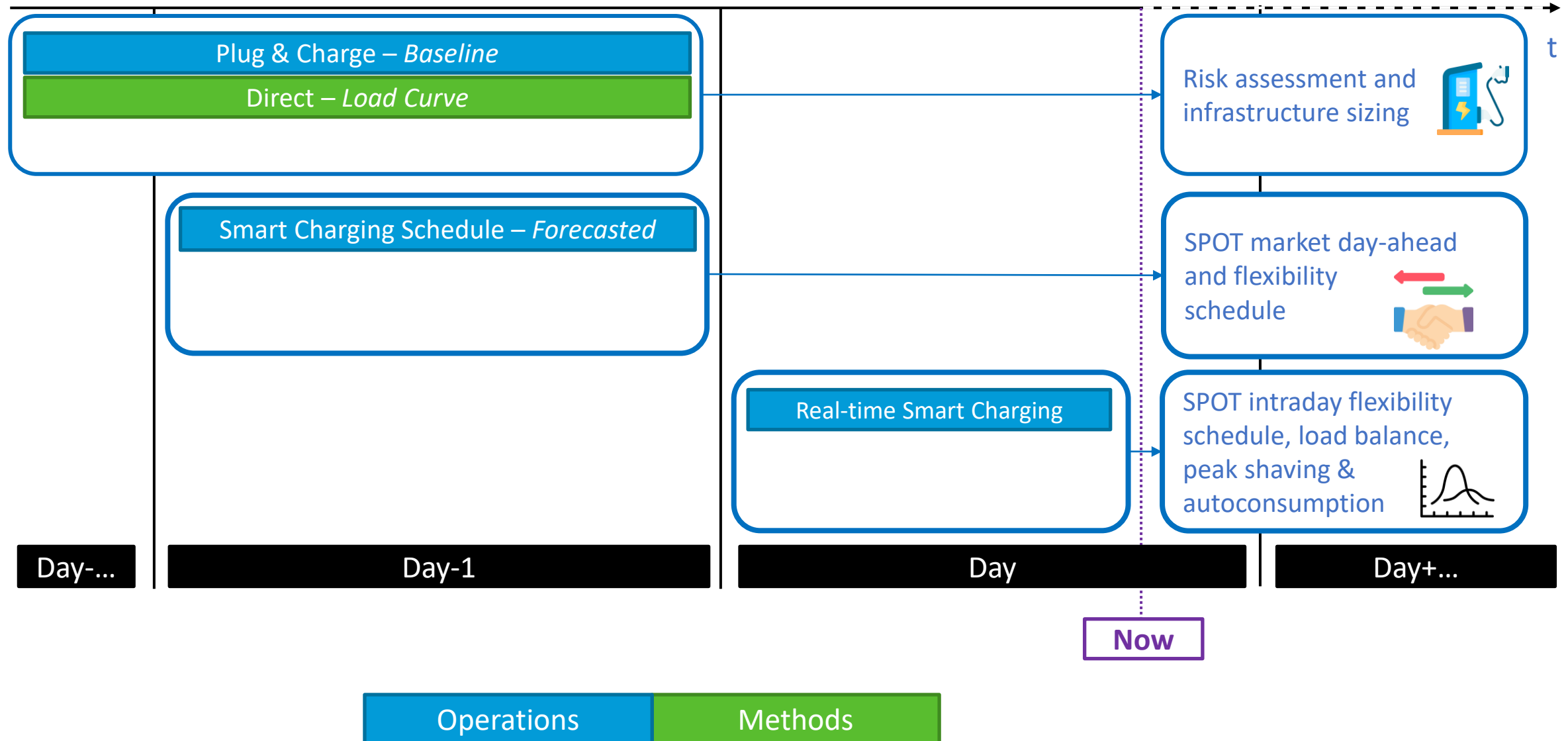
Accurately anticipating incoming EV charging demand is not enough, we need some sense of the uncertainty around the predictions to accommodate for different scenarios.



Operations and methods across the timeline



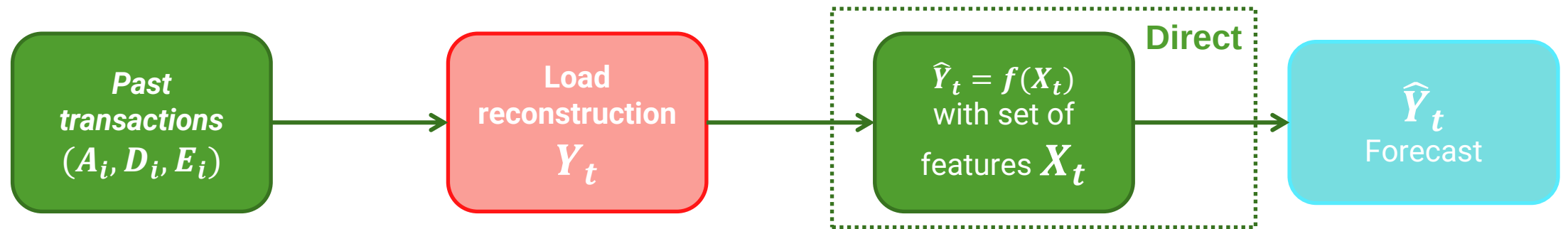
Operations and methods across the timeline



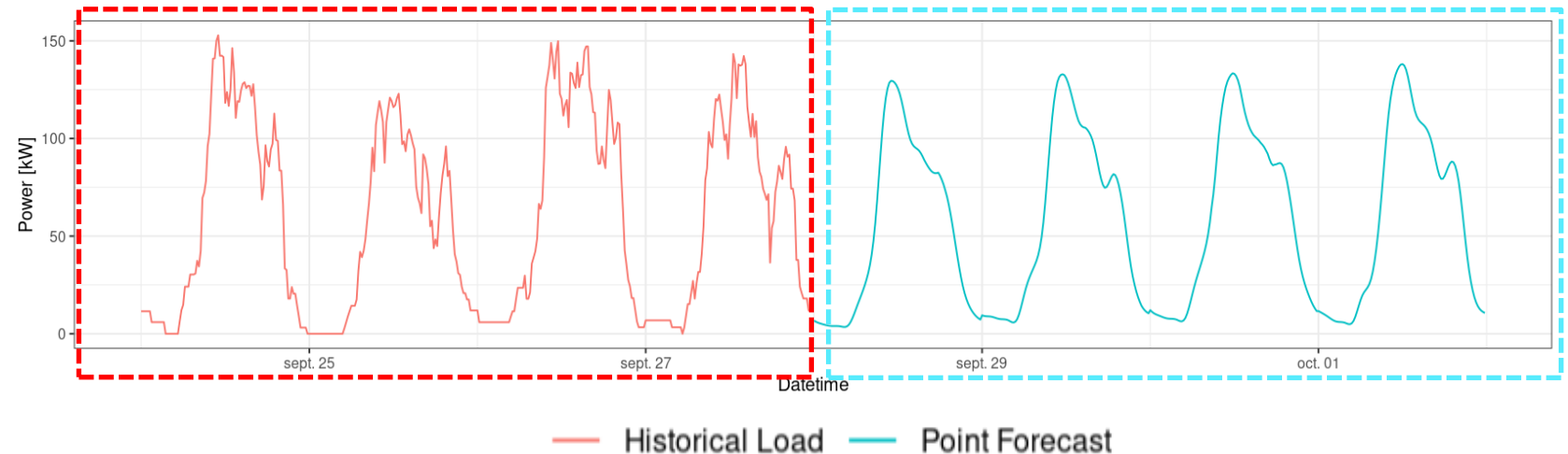
Direct approaches

Interests

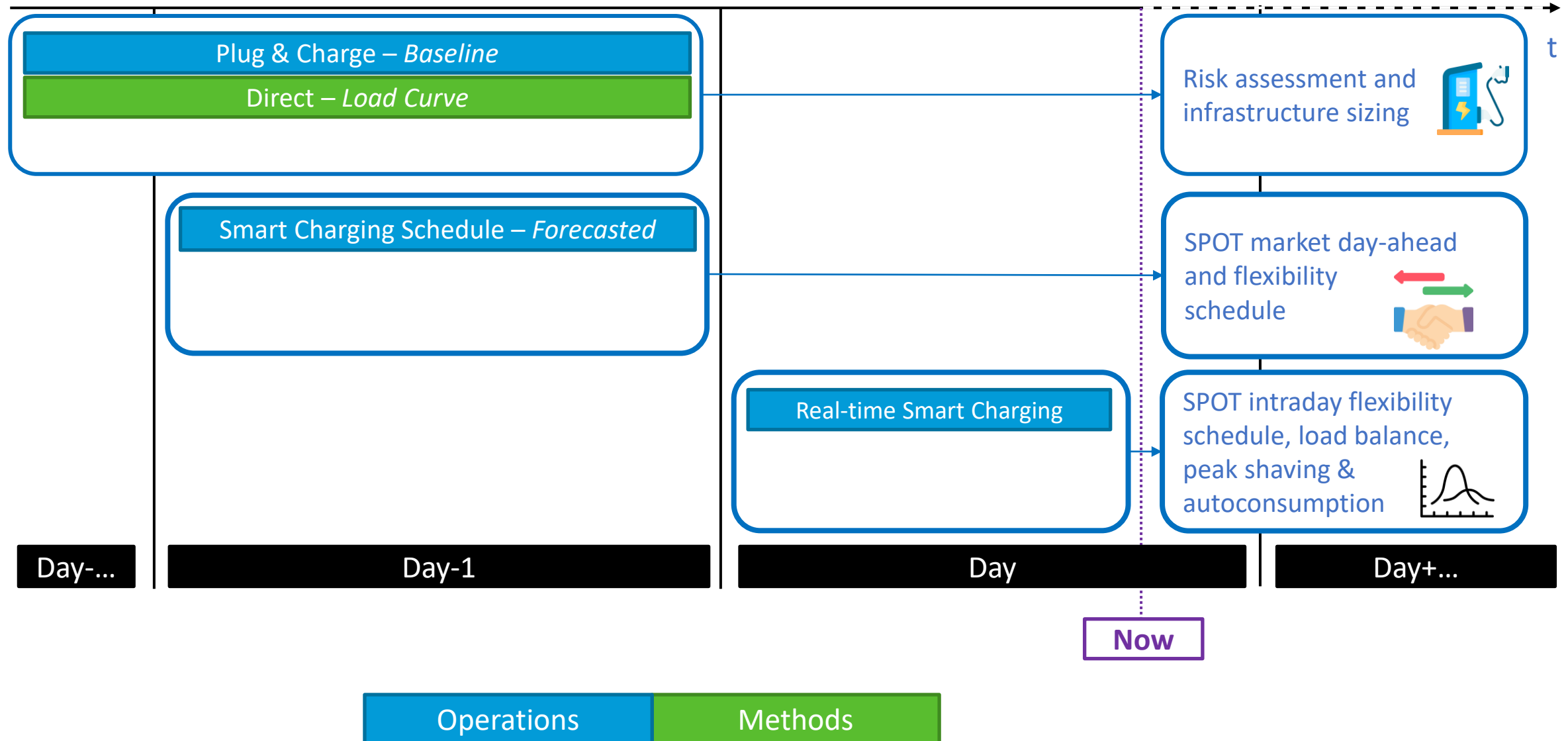
- Plug & Charge load forecasting
- Baseline to which other approaches can be compared



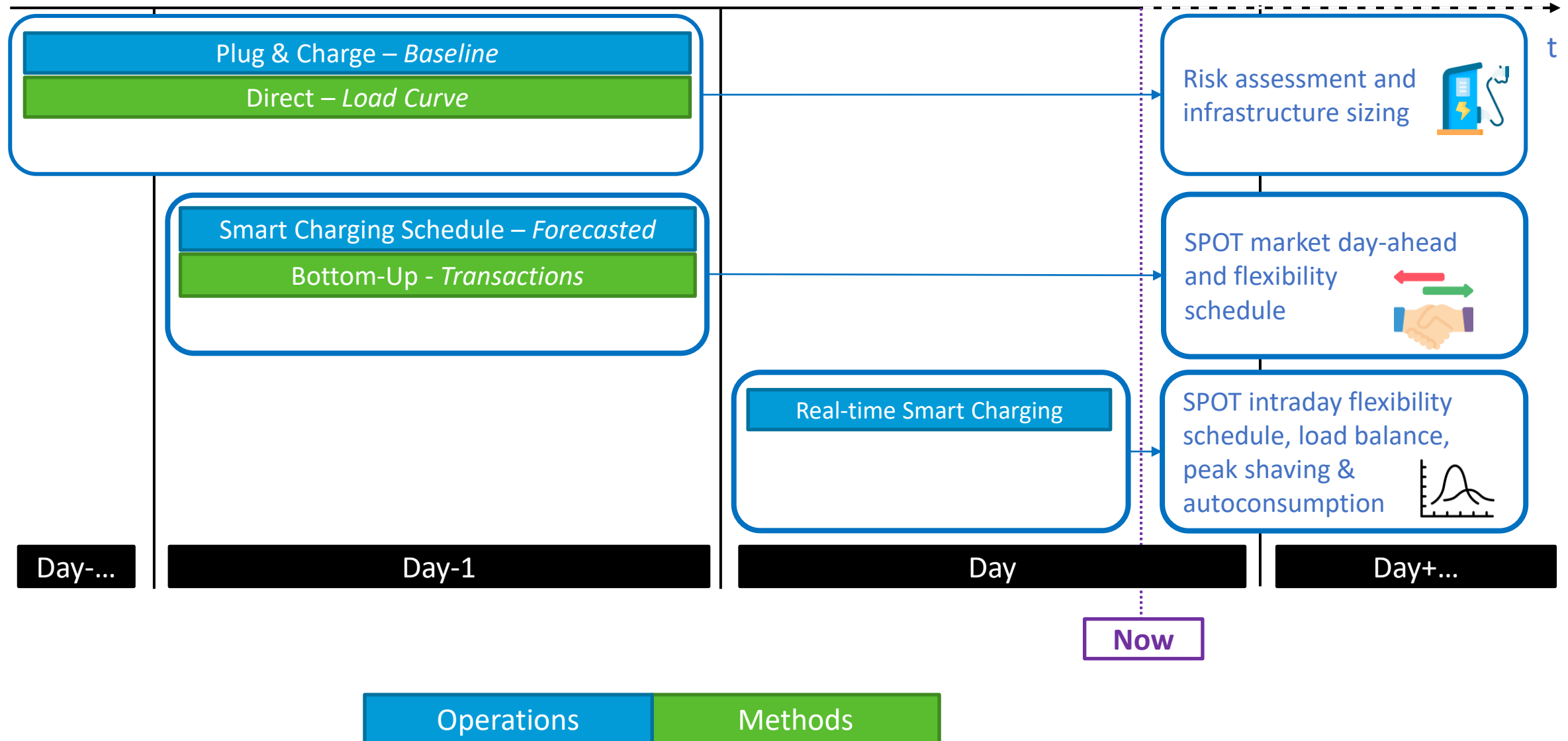
i	A_i	D_i	E_i
1	9h	5h	7kW h
2	12h	2h	3kW h
...	...	...	...
N	17h	1h	1kW h



Operations and methods across the timeline



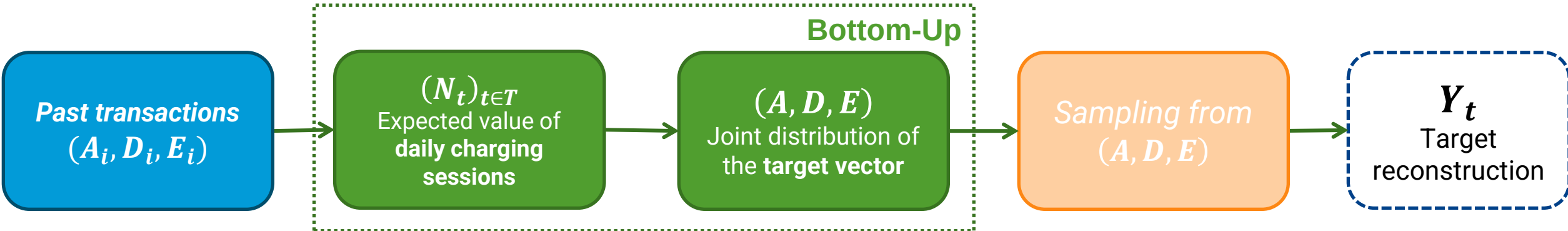
Operations and methods across the timeline



Bottom-Up approaches

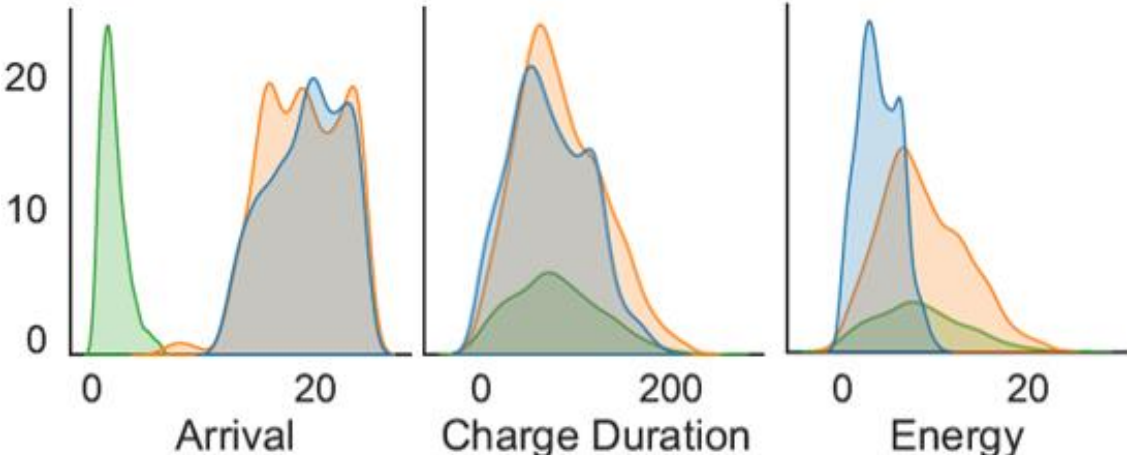
Interests

- **Finer granularity of forecasting (at the transaction level)**
- **Allows smart charging applications**
 - More leverage to optimise at the individual curve level
 - Optimisation on intra-day markets
 - Self-consumption and V2H applications



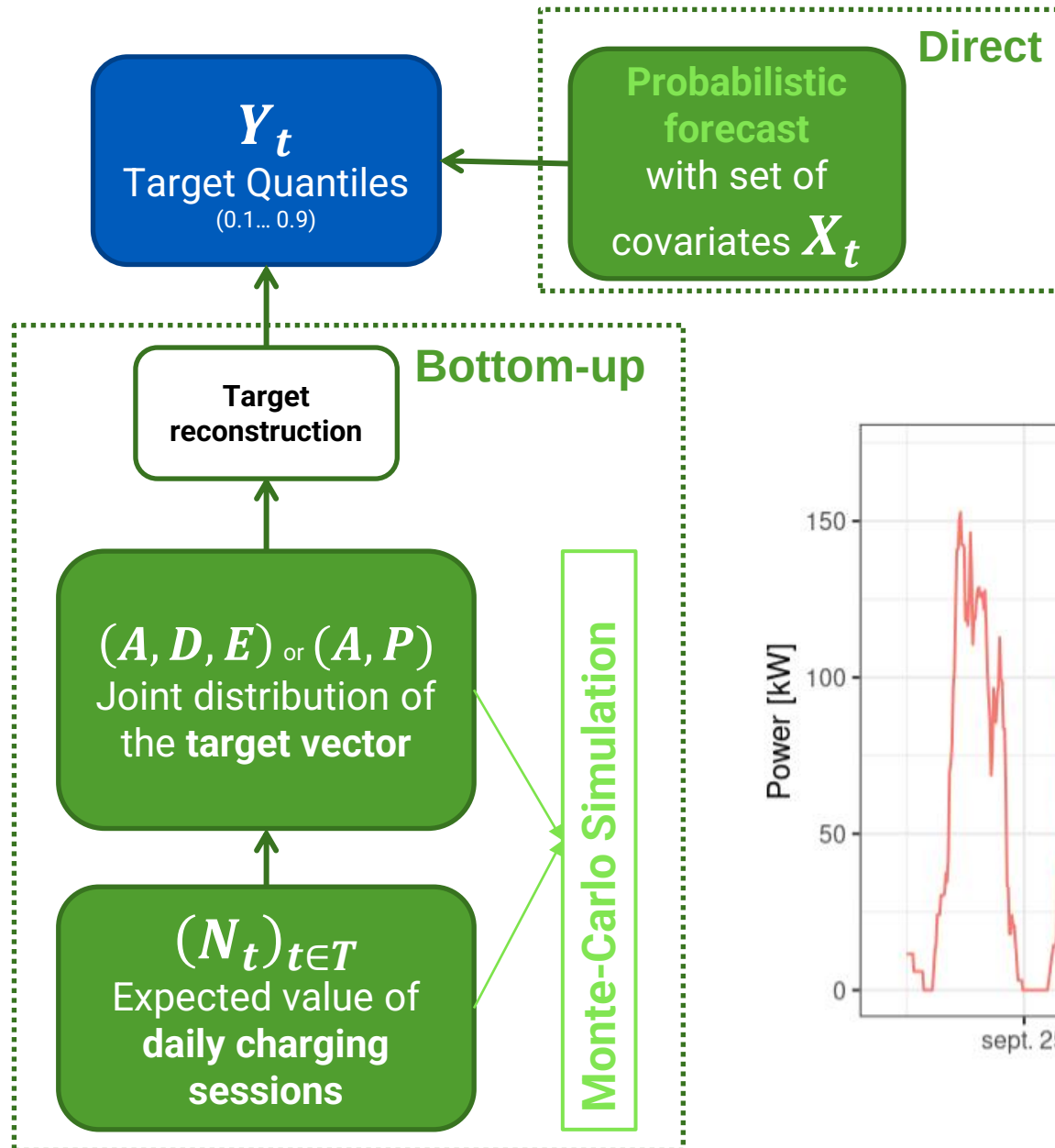
Main clusters

i	A_i	D_i	E_i
1	9h	5h	7kWh
2	12h	2h	3kWh
...	...	...	...
N	17h	1h	1kWh



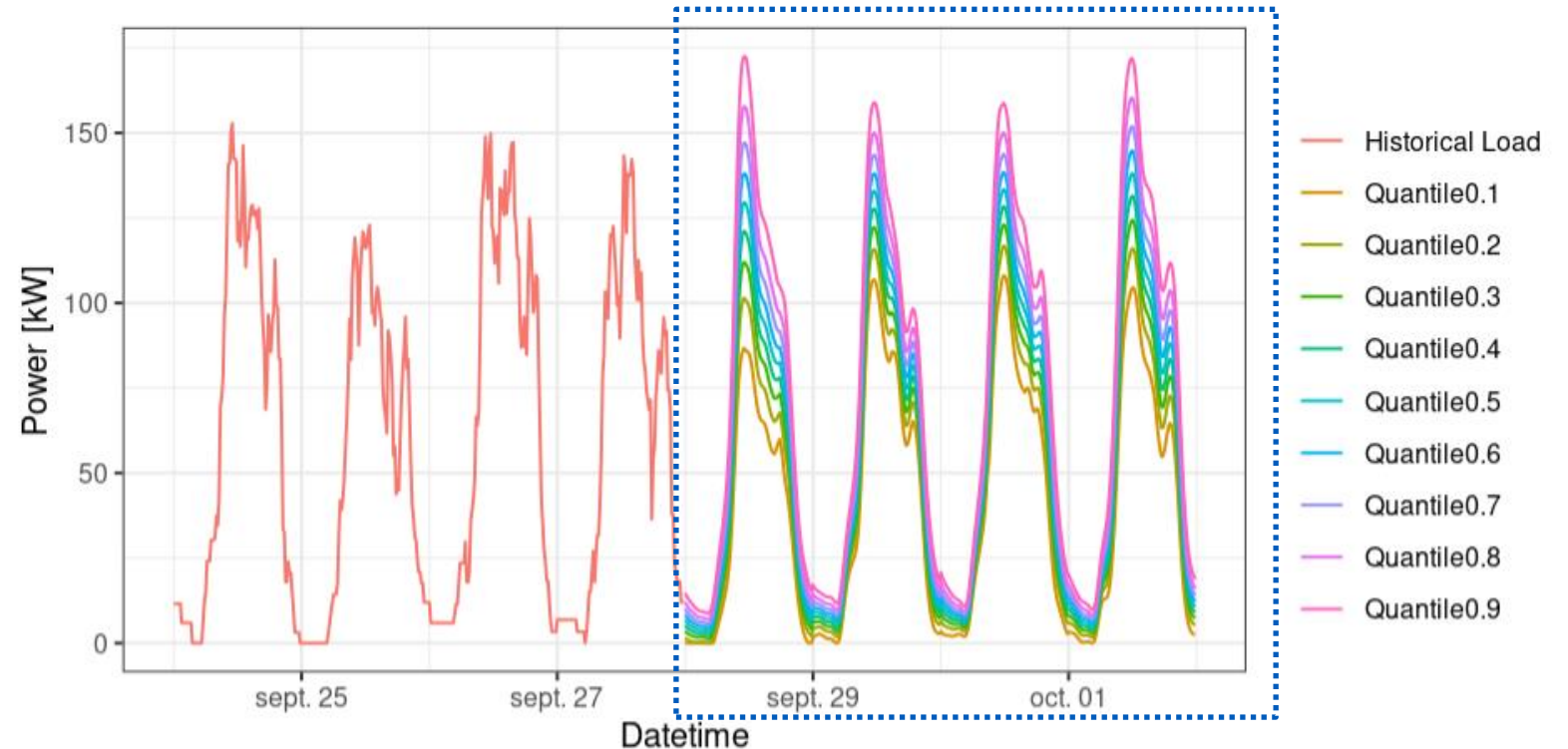
i	A_i	D_i	E_i
1	8h	4h	9kWh
2	11h	1h	1kWh
...	...	...	...
$\hat{N}$	13h	2h	1kWh

From point forecasts to probabilistic forecasts

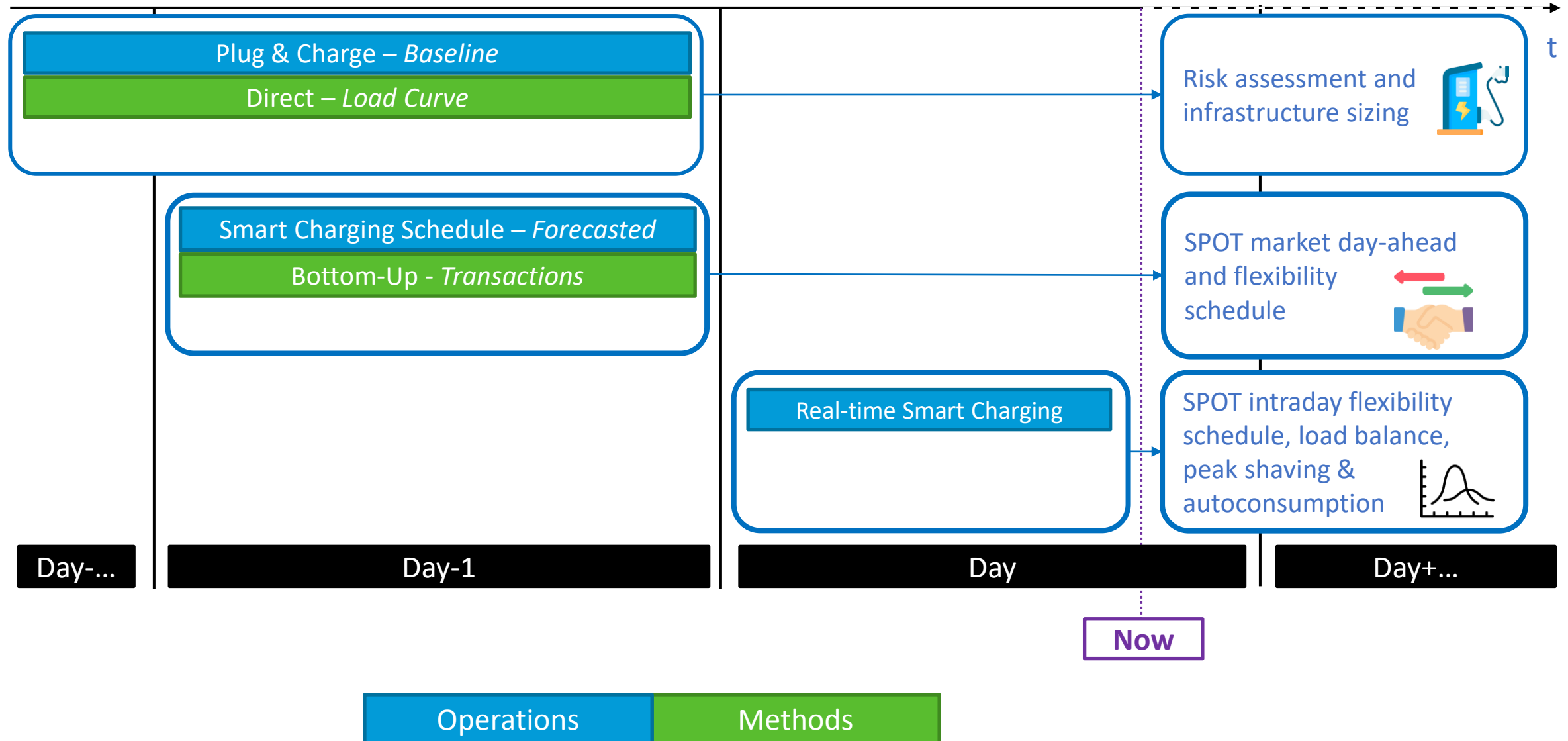


Interests

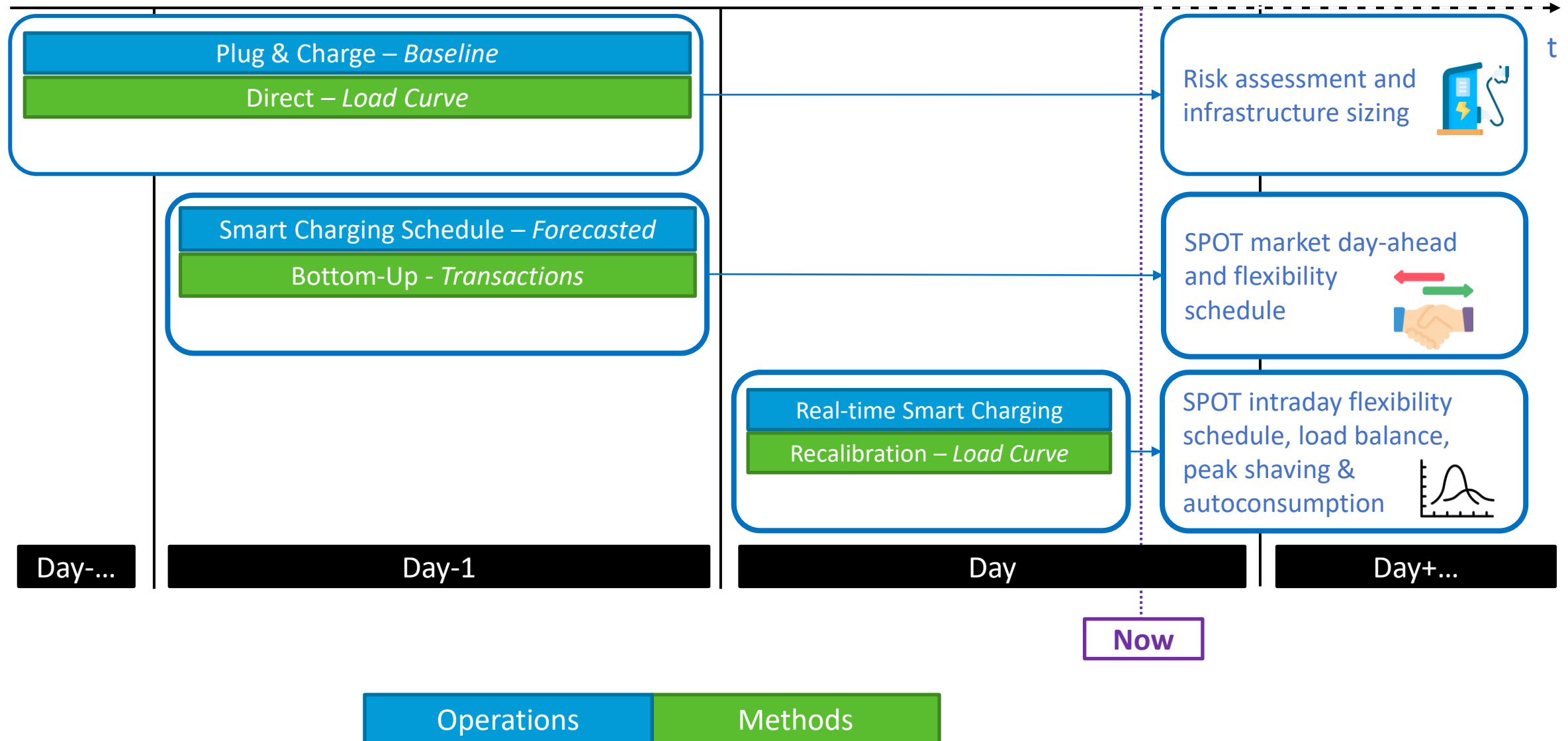
- Need to quantify the uncertainty around these forecasts to assess the risk around the smart charging operations
- The optimisation is different if there is an important uncertainty around the expected load (higher energy, shorter durations etc.)



Operations and methods across the timeline



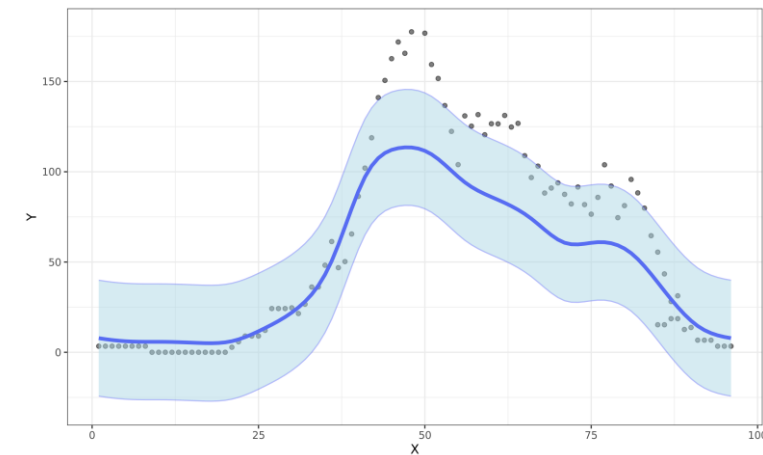
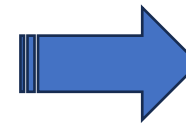
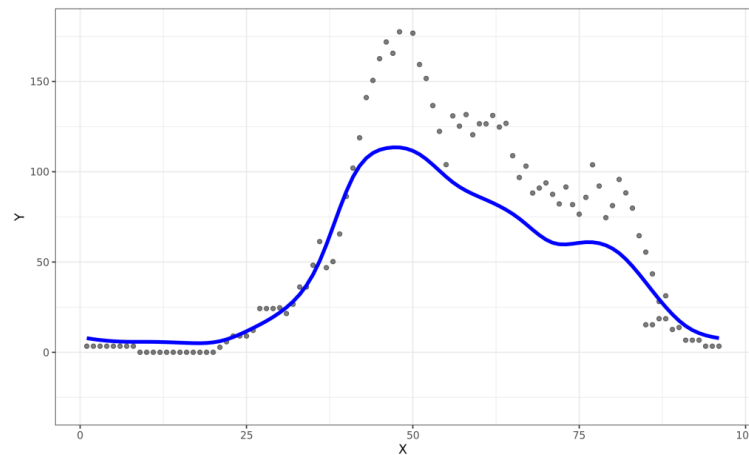
Operations and methods across the timeline



Theoretical Focus on Conformal Prediction

- Introduction to Split Conformal Prediction (SCP)
- Extension to Time Series with Adaptive Conformal Inference (ACI)

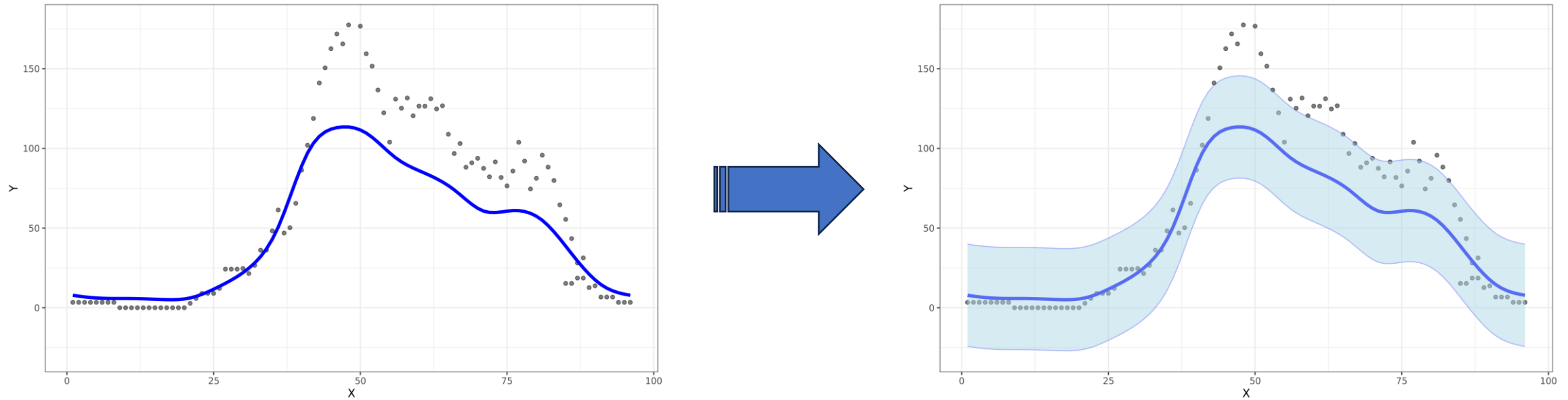
- Turns a point forecast into a prediction set



- Corrects a miscalibrated prediction set

Motivation of Conformal Prediction (CP)

- Turns a point forecast into a prediction set



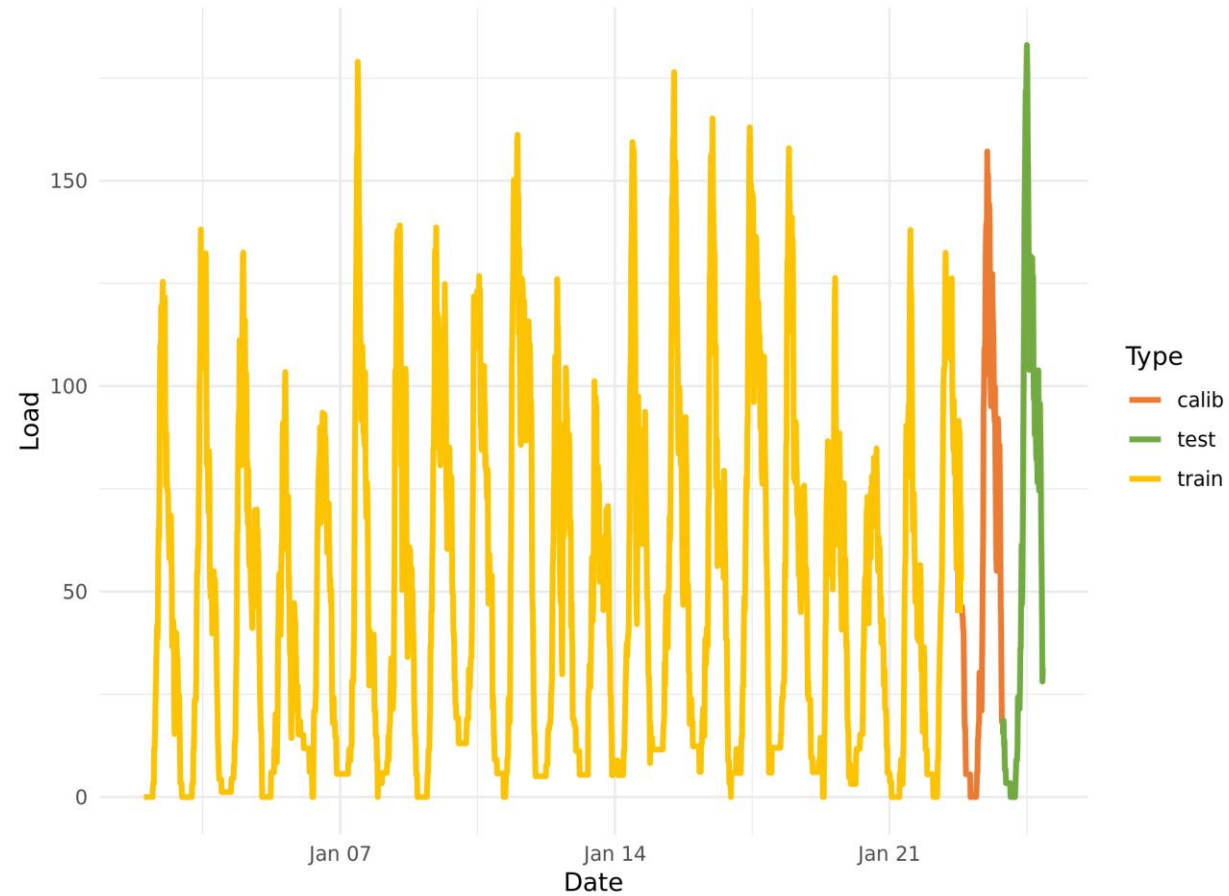
- Corrects a miscalibrated prediction set

Split Conformal Prediction (SCP)

Training

Calibration

Test

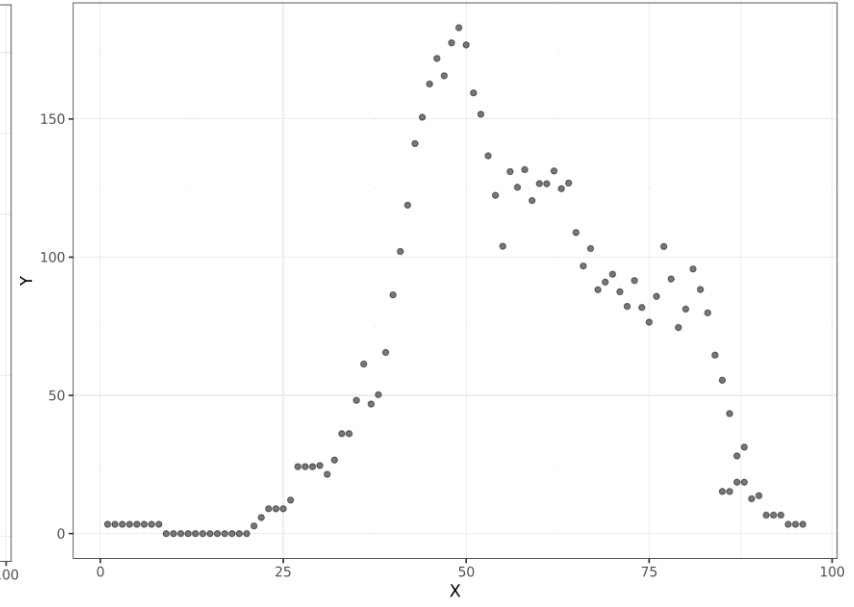
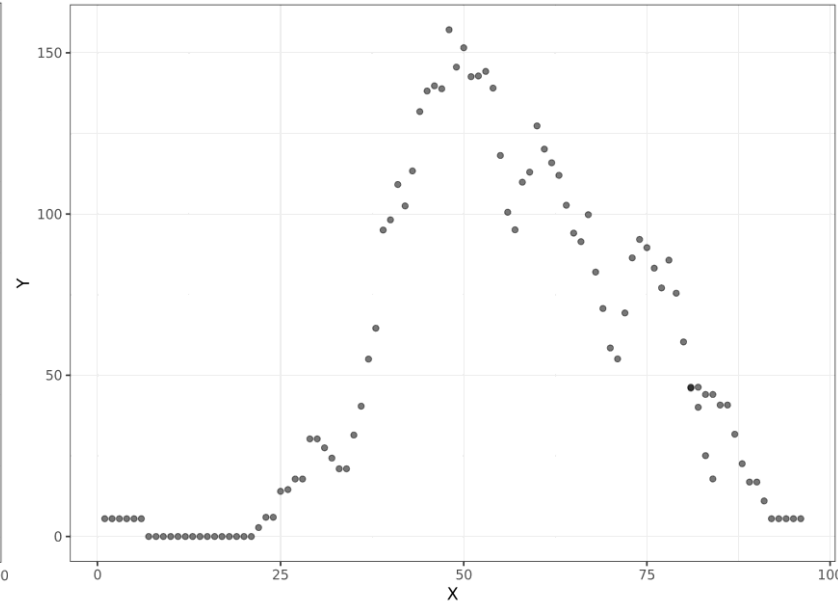
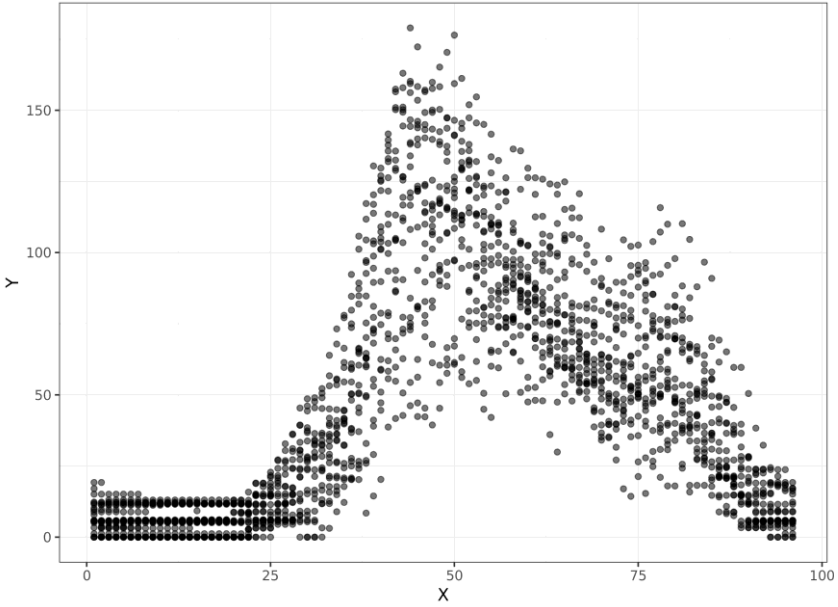


Split Conformal Prediction (SCP)

Training

Calibration

Test

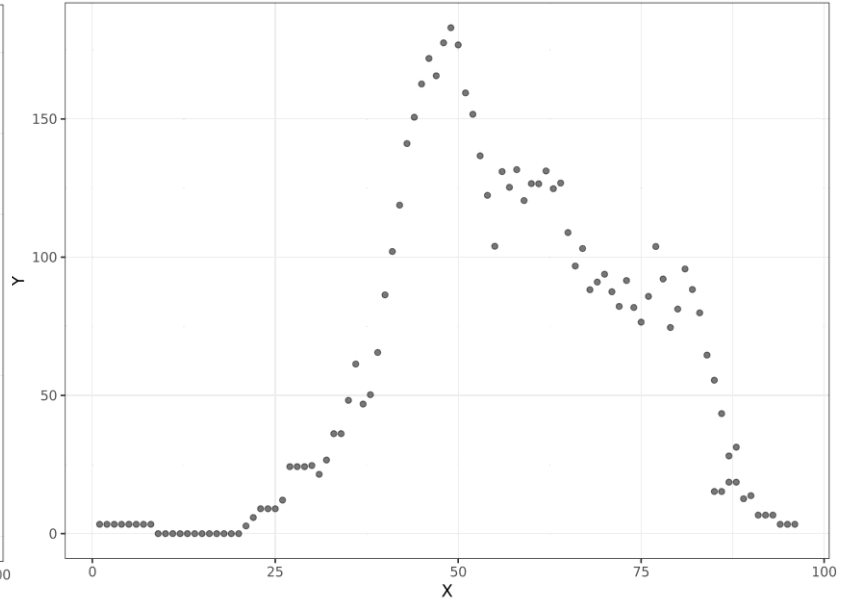
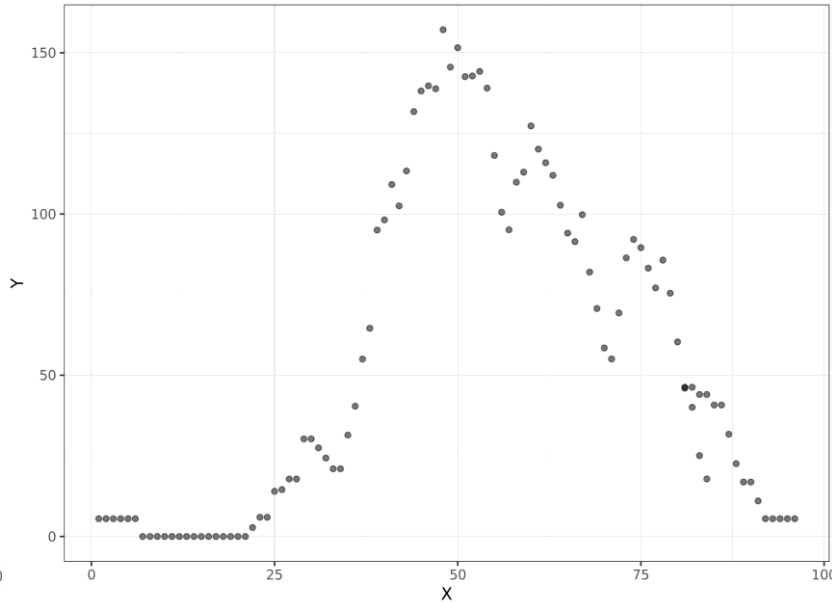
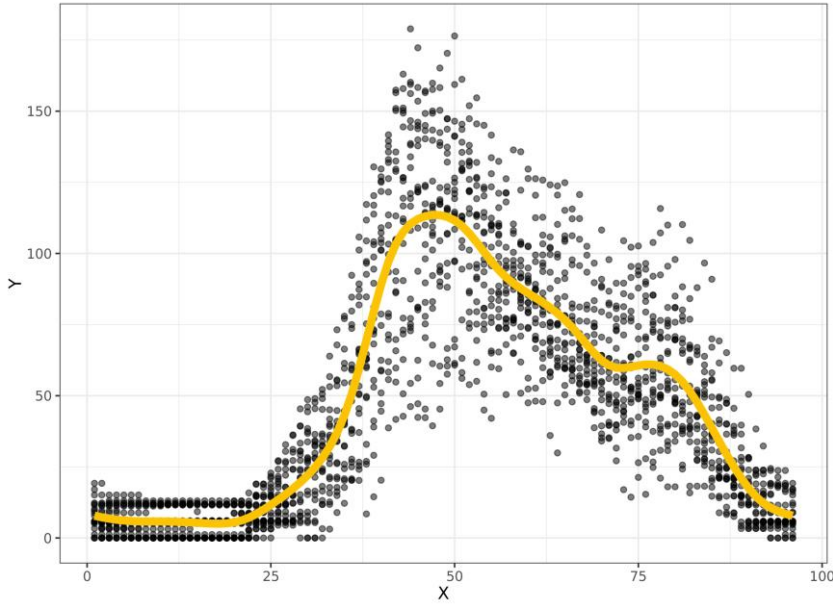


Split Conformal Prediction (SCP)

Training

Calibration

Test



— X, Y —→

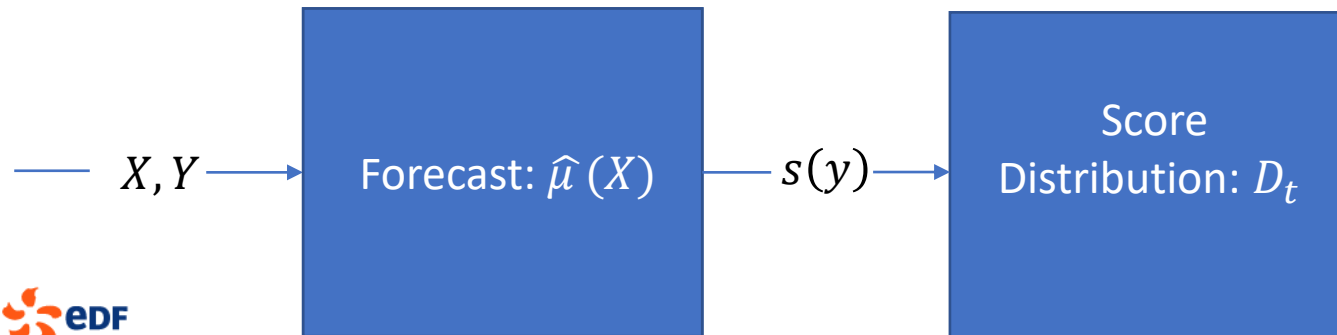
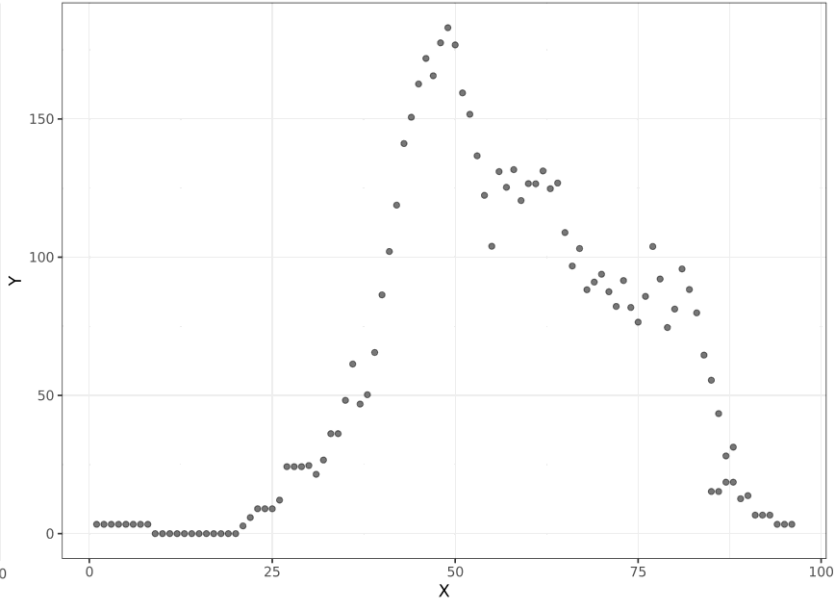
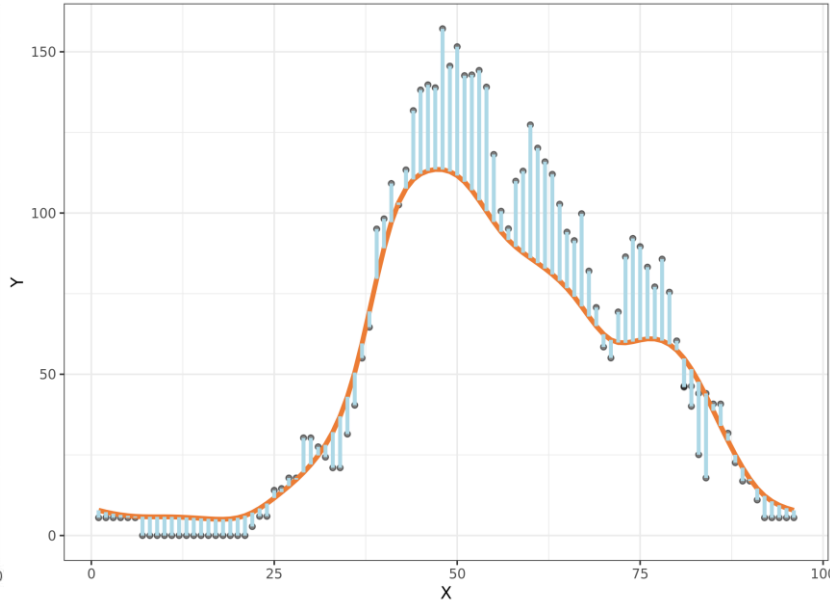
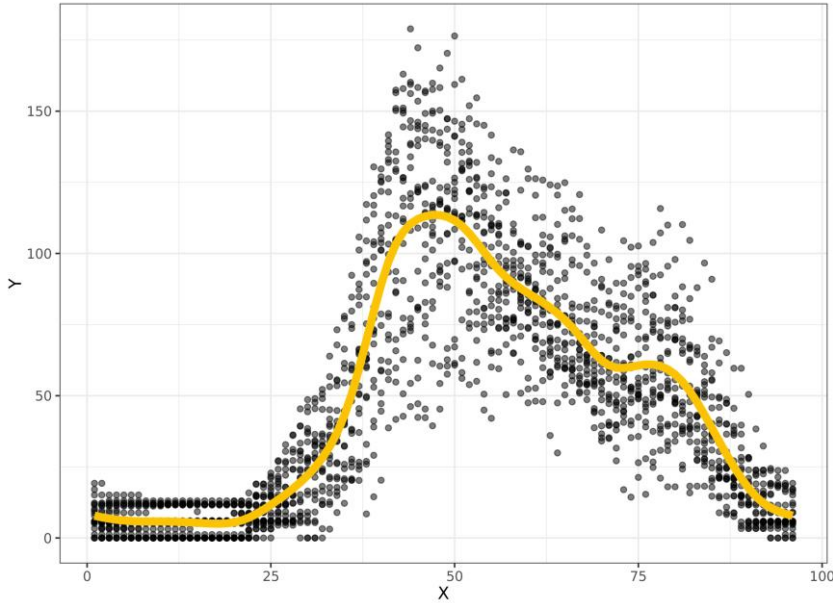
Forecast: $\hat{\mu}(X)$

Split Conformal Prediction (SCP)

Training

Calibration

Test

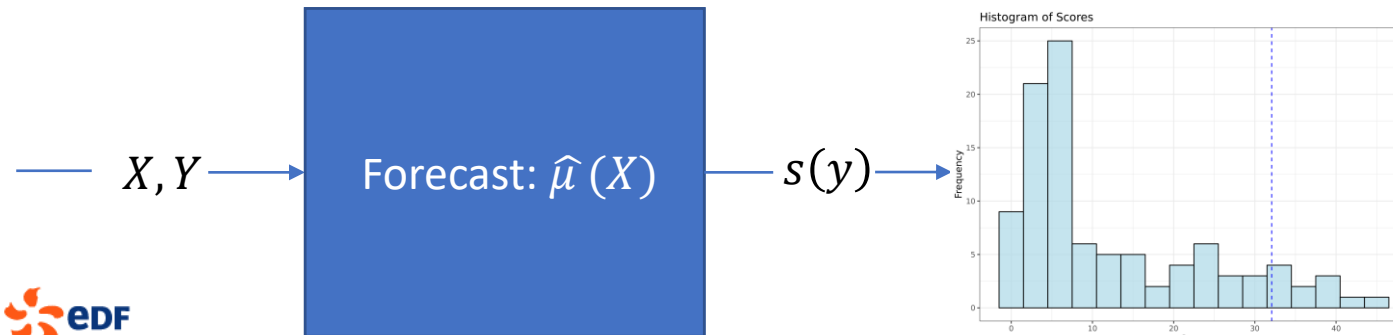
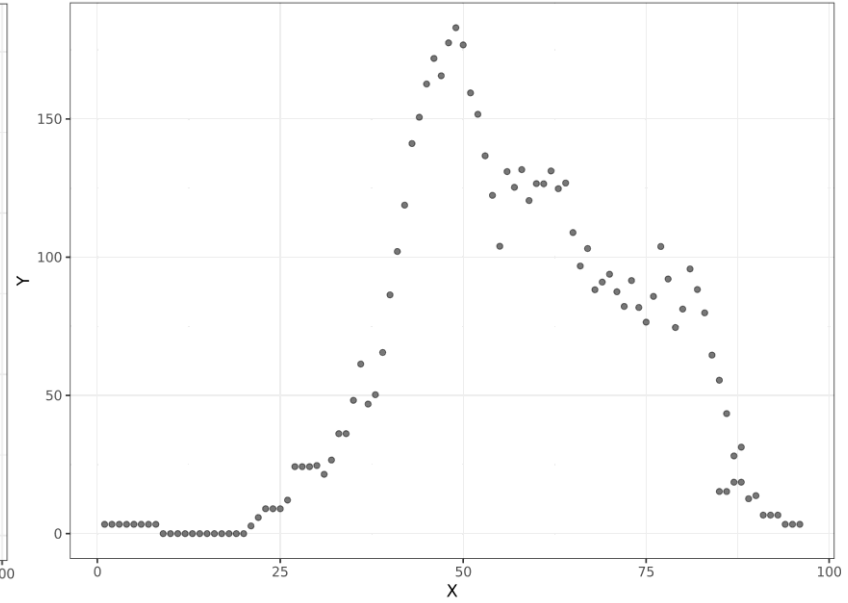
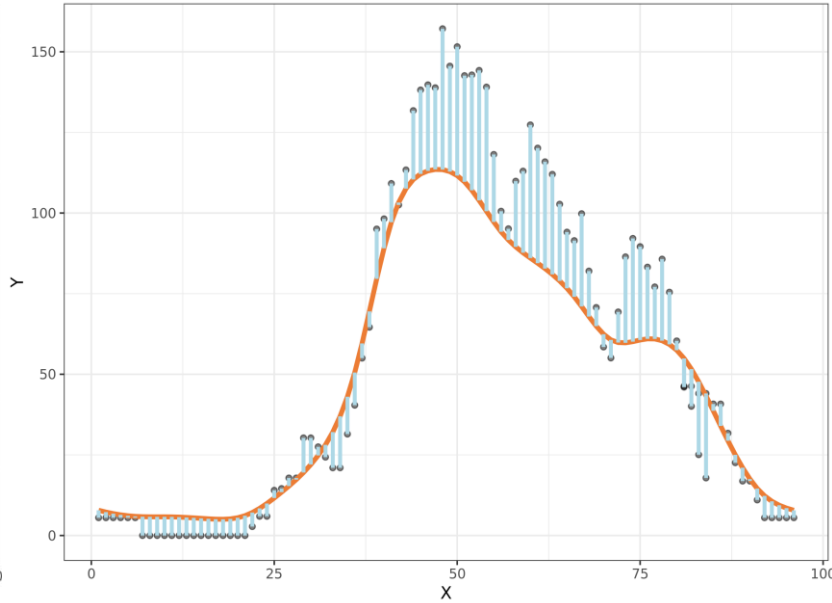
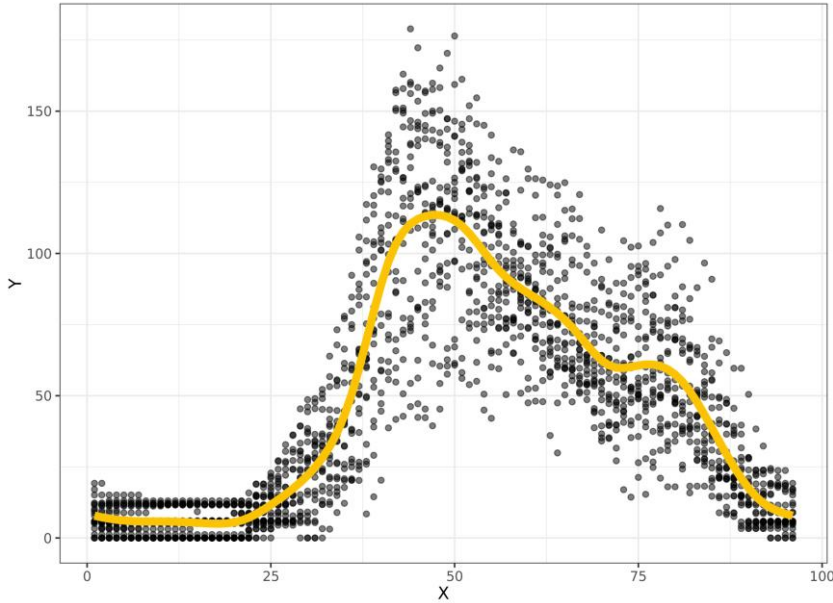


Split Conformal Prediction (SCP)

Training

Calibration

Test

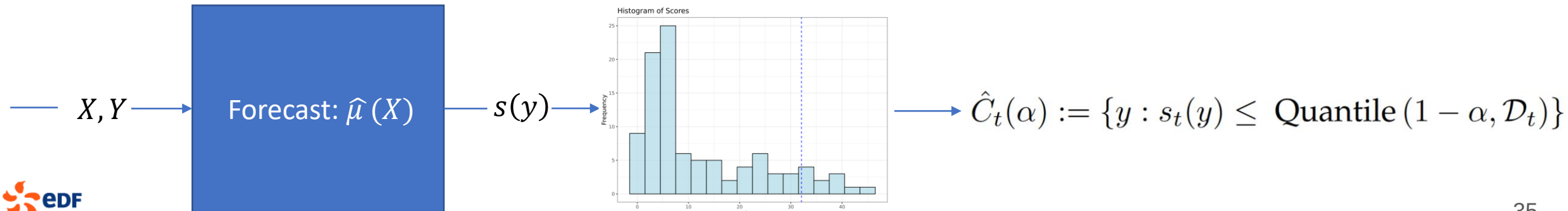
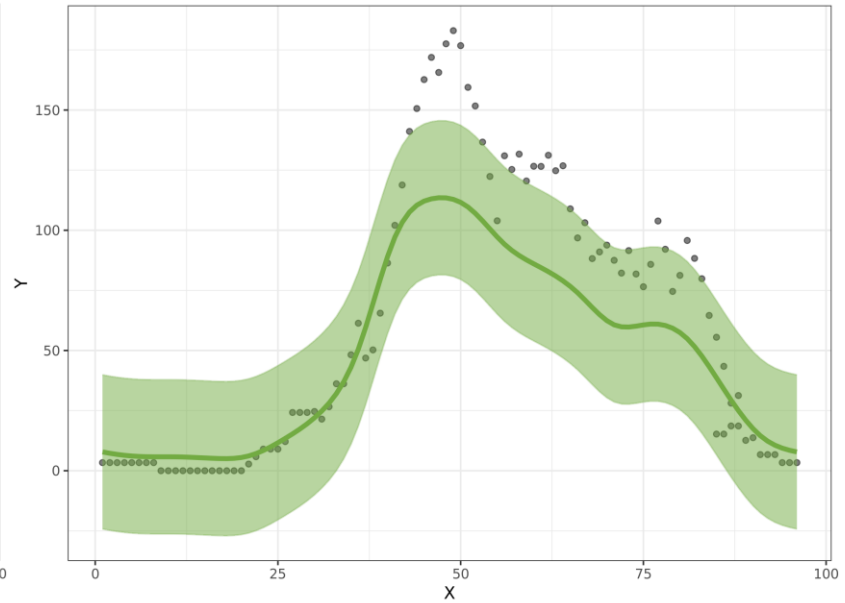
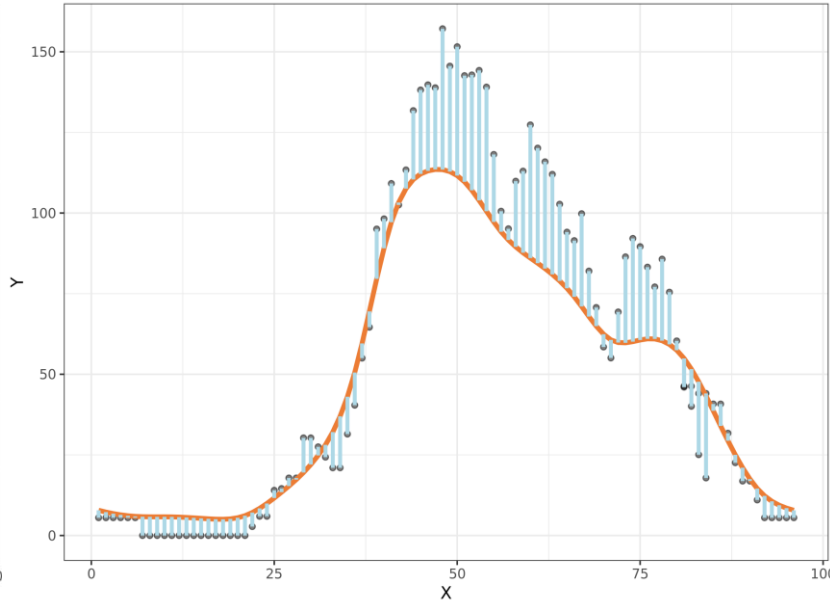
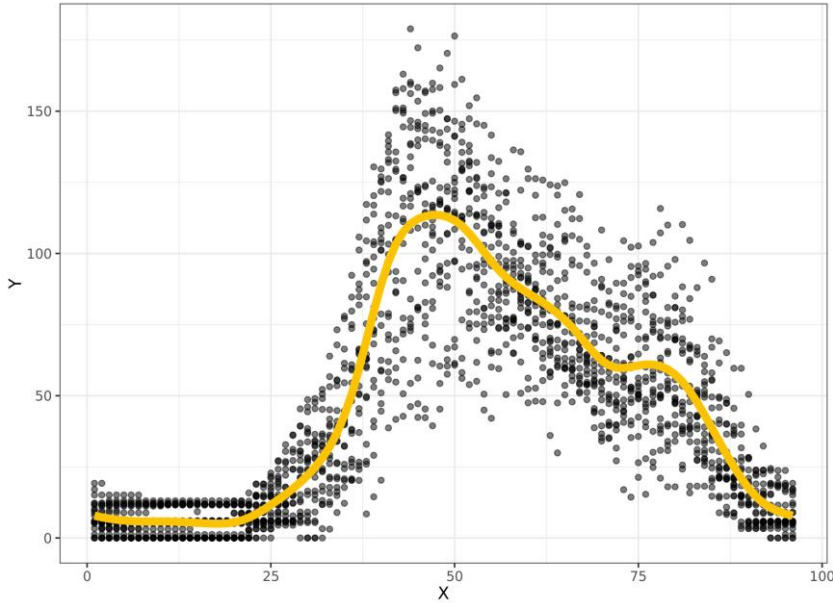


Split Conformal Prediction (SCP)

Training

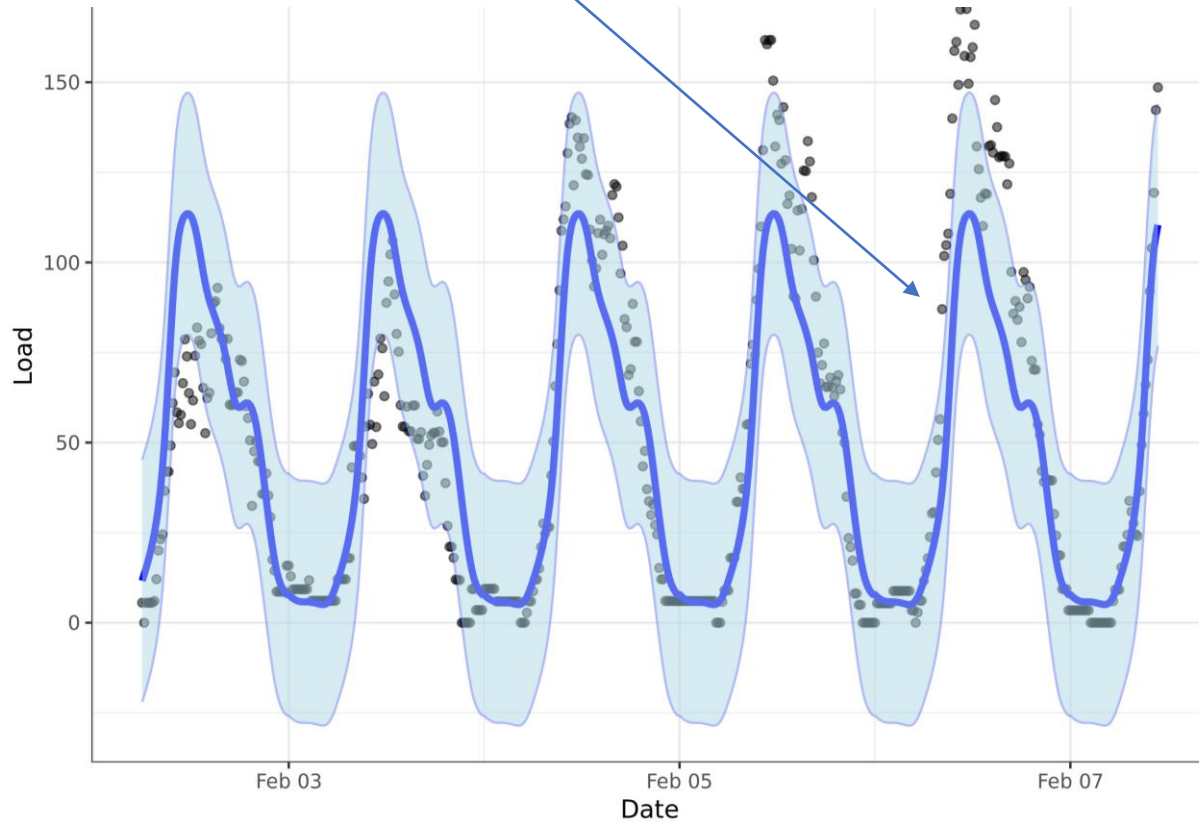
Calibration

Test

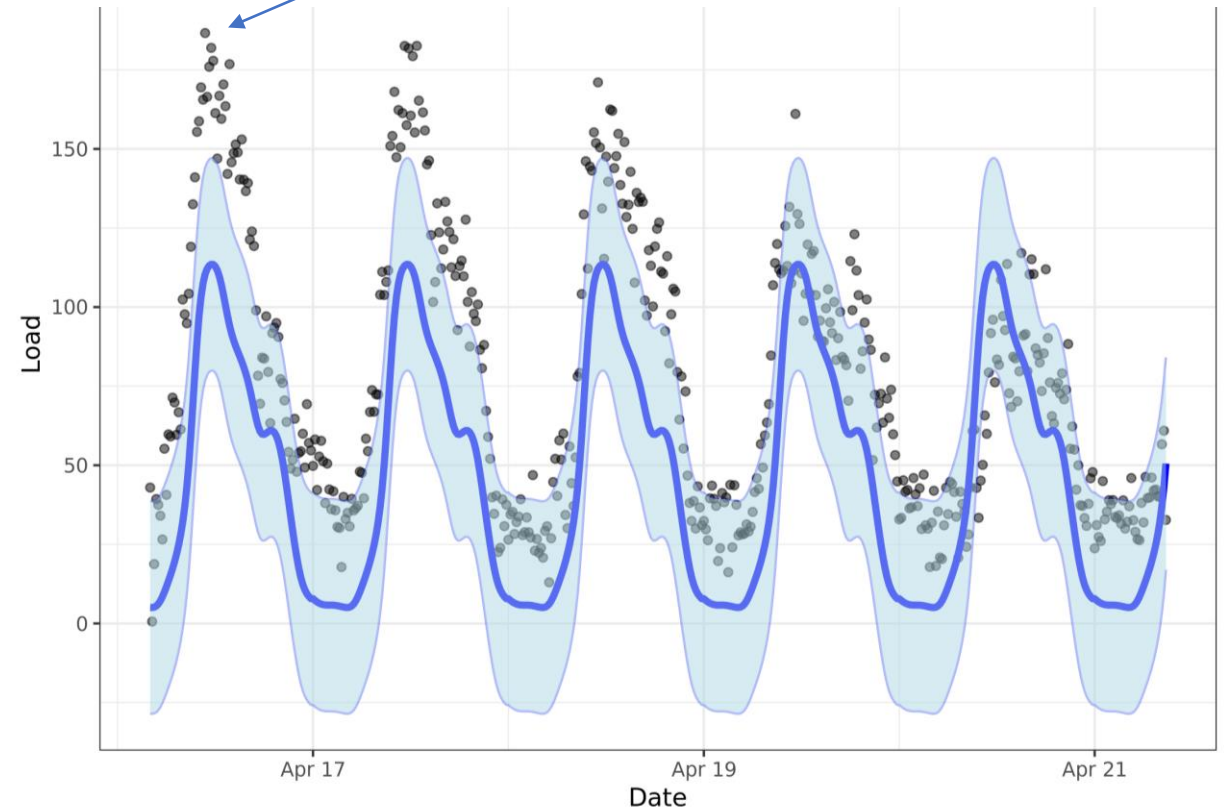


The Time Series setting

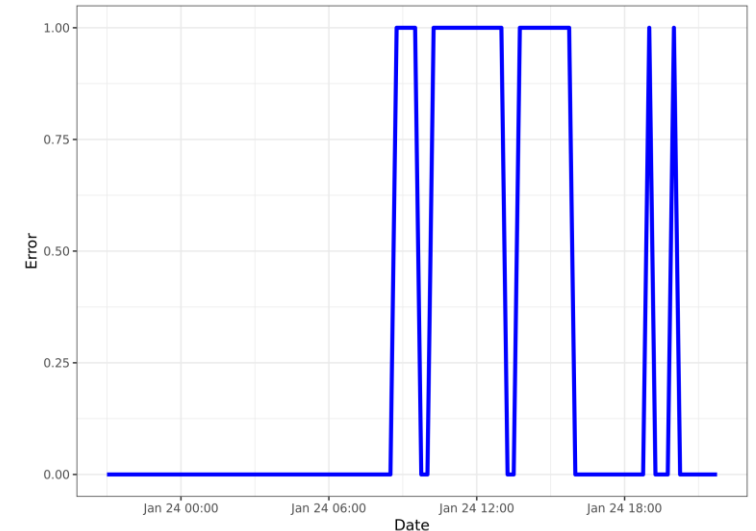
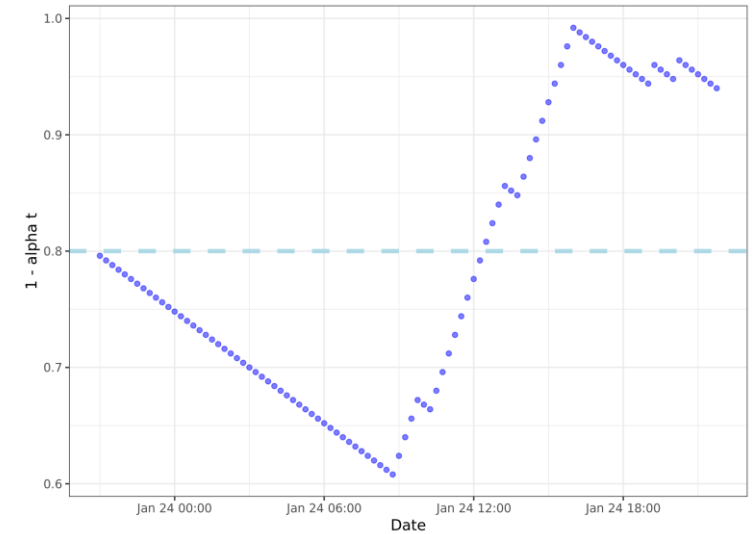
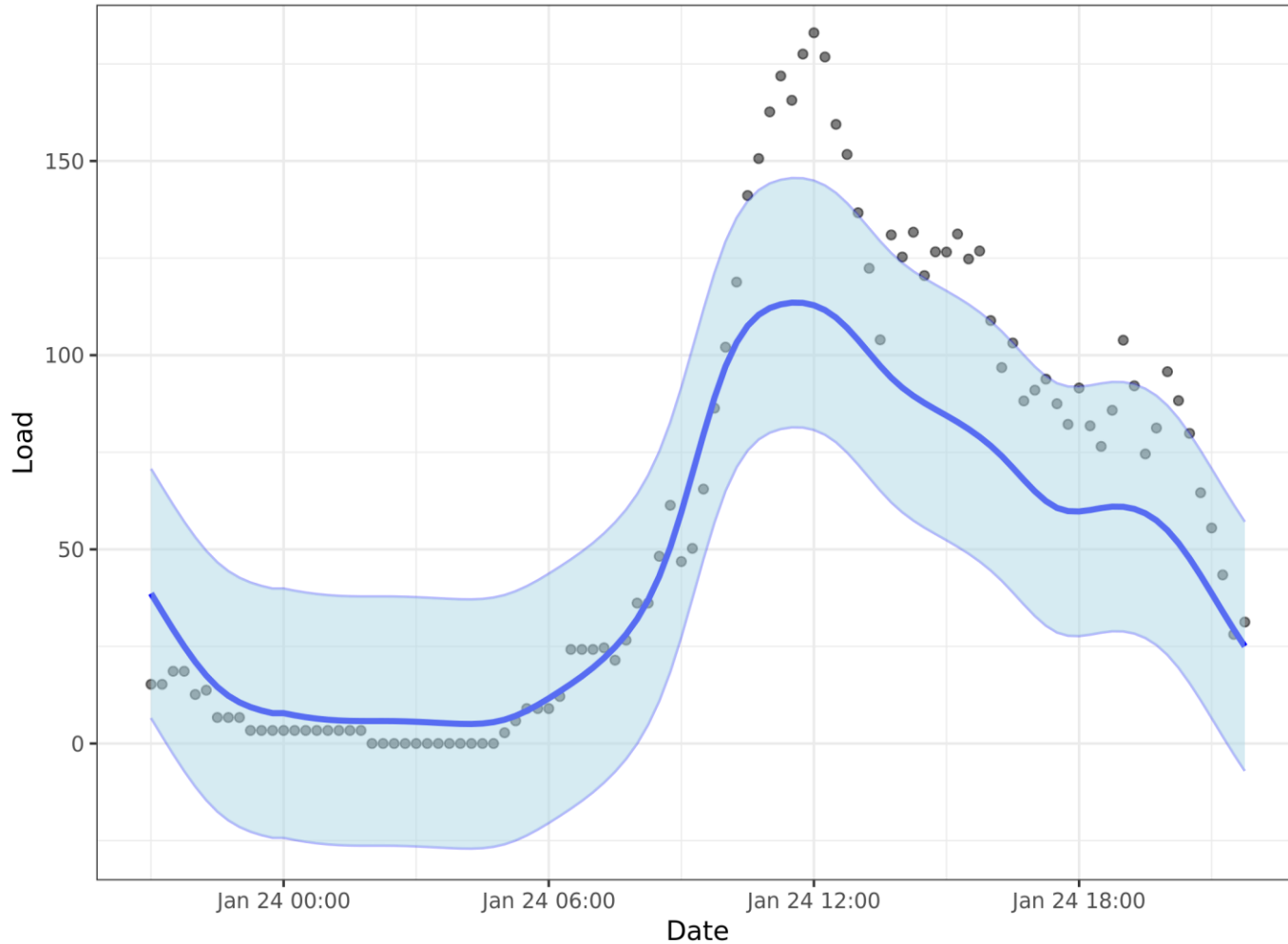
Dependency in high consumption events



Drift both in term of Mean and Deviation

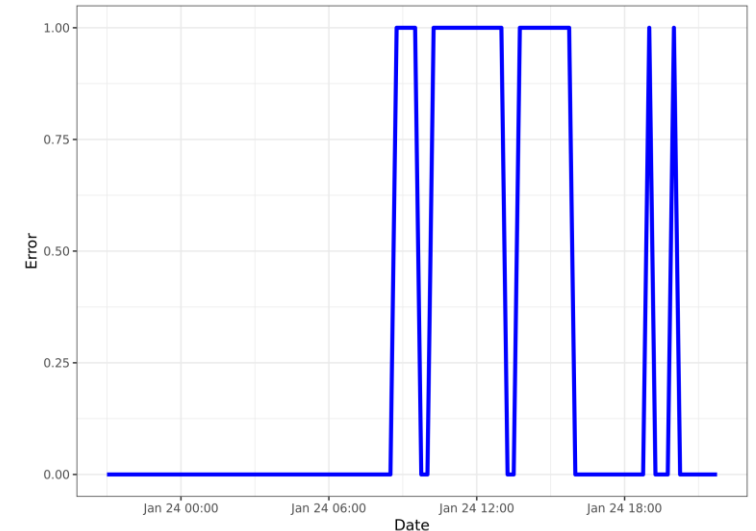
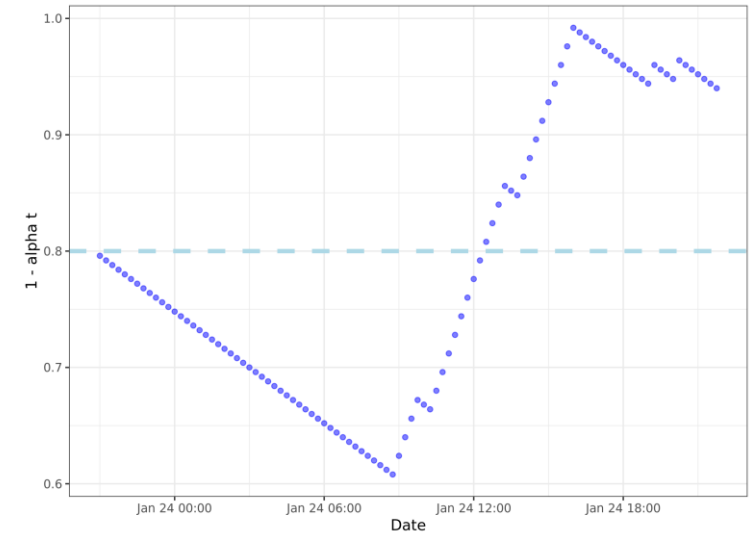
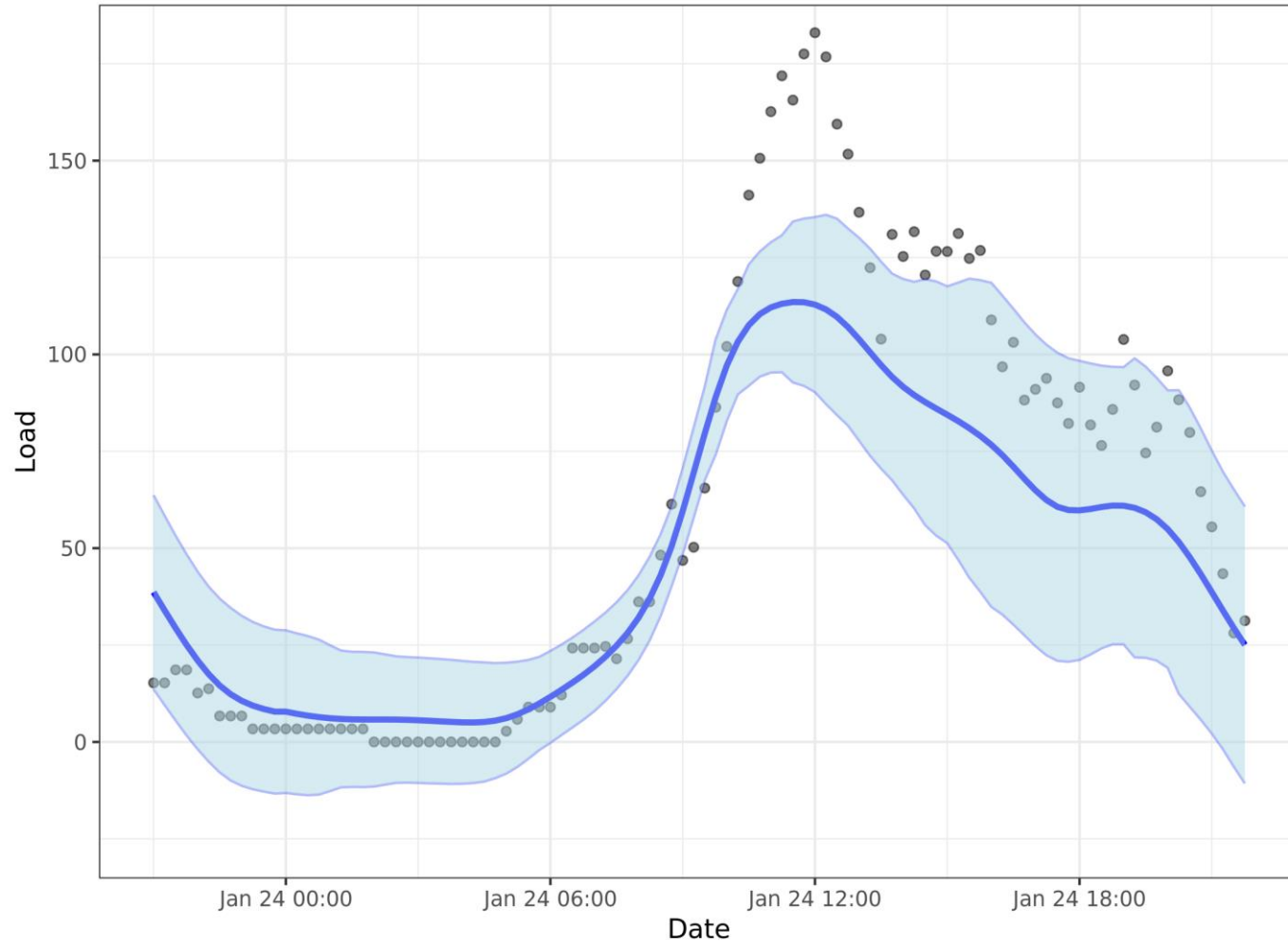


Adaptive Conformal Inference: Gibbs & Candès (2021)



$$\alpha_{t+1} = \alpha_t + \gamma (\alpha - \text{err}_t) \quad \text{with } \text{err}_t = \mathbb{1}_{\{Y_t \notin \hat{C}_t^\alpha\}}$$

Adaptive Conformal Inference: Gibbs & Candès (2021)



$$\alpha_{t+1} = \alpha_t + \gamma (\alpha - \text{err}_t) \quad \text{with } \text{err}_t = \mathbb{1}_{\{Y_t \notin \hat{C}_t^\alpha\}}$$

How to evaluate the quality of a prediction set ?

Validity:

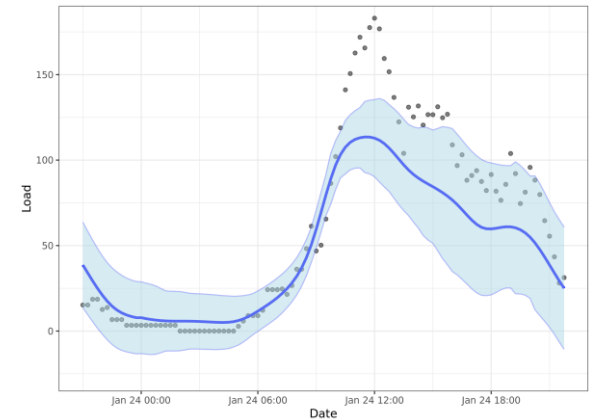
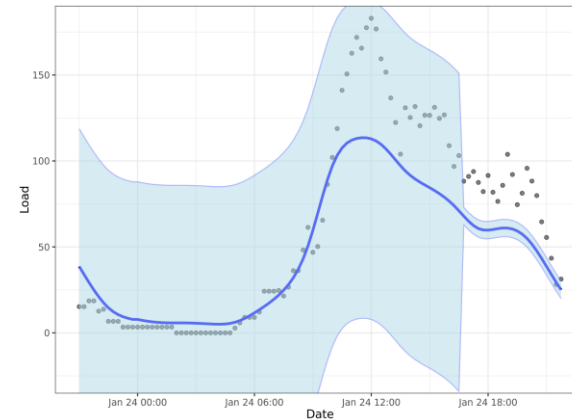
➤ Probabilistic:

$$\mathbb{P} \left(Y_{test} \in \hat{C}_{test} \right) \geq 1 - \alpha$$

➤ Asymptotic:

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathbb{1}_{\{Y_t \in \hat{C}_t\}} \stackrel{a.s.}{=} 1 - \alpha$$

Efficiency:



➤ Interval Length

Coverage guarantees and sketch of proof

$$\alpha_{t+1} = \alpha_t + \gamma (\alpha - \text{err}_t)$$

$$\frac{1}{T} \sum_{t=1}^T (\alpha - \text{err}_t) = \frac{\alpha_{T+1} - \alpha_1}{T\gamma} \quad \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathbb{1}_{\{Y_t \in \hat{C}_t\}} \stackrel{a.s.}{=} 1 - \alpha$$

- This proof only works with constant γ
- There is a tradeoff with the choice of γ . The higher it is the faster we get the coverage but the more instable the algorithm is.

How to choose the gradient step size ?

- We want to maintain the validity while being more efficient
- To tackle this issue, we need to use a **dynamic step size**:

¹Zaffran et al. (2022)

²Gibbs & Candès (2022)

³Bhatnagar et al. (2023)

⁴Angelopoulos et al. (2024)

My presentation at JDS (2024)

Meta algorithm to obtain γ_t

Explicit γ_t

¹Adaptive conformal predictions for time series / ²Conformal inference for online prediction with arbitrary distribution shifts / ³Improved online conformal prediction via strongly adaptive online learning / ⁴Online conformal prediction with decaying step sizes

The Decaying step-size

➤ Formula based on regret minimization: Orabona (2019) $\gamma_t = \frac{\eta}{\sqrt{\sum_{s=1}^t \|g_s\|^2}}$

$$\alpha_{t+1} = \alpha_t + \frac{\eta}{\sqrt{\sum_{s=1}^t (\text{err}_s - \alpha)^2}} (\alpha - \text{err}_t)$$


Asymptotically Valid

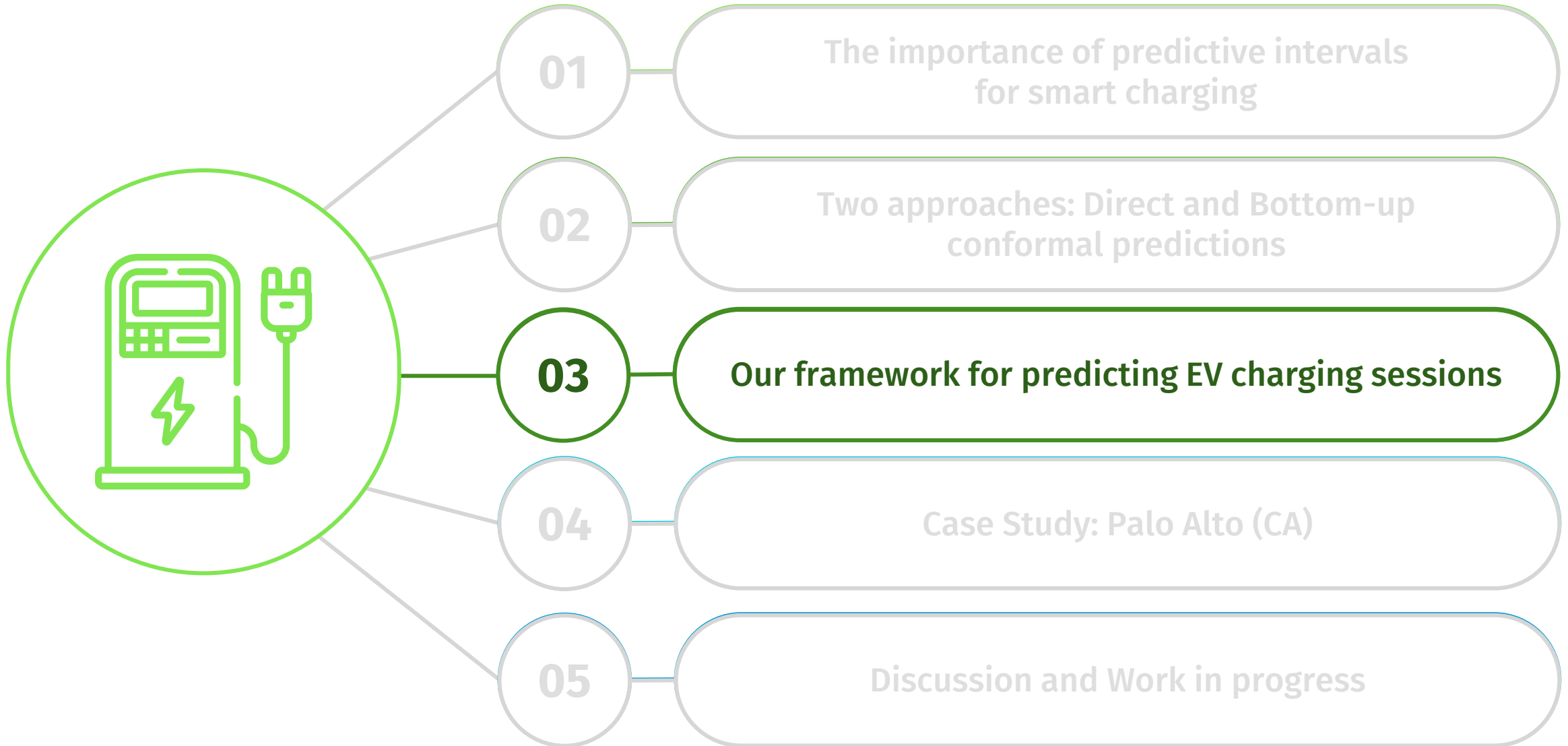


Efficient

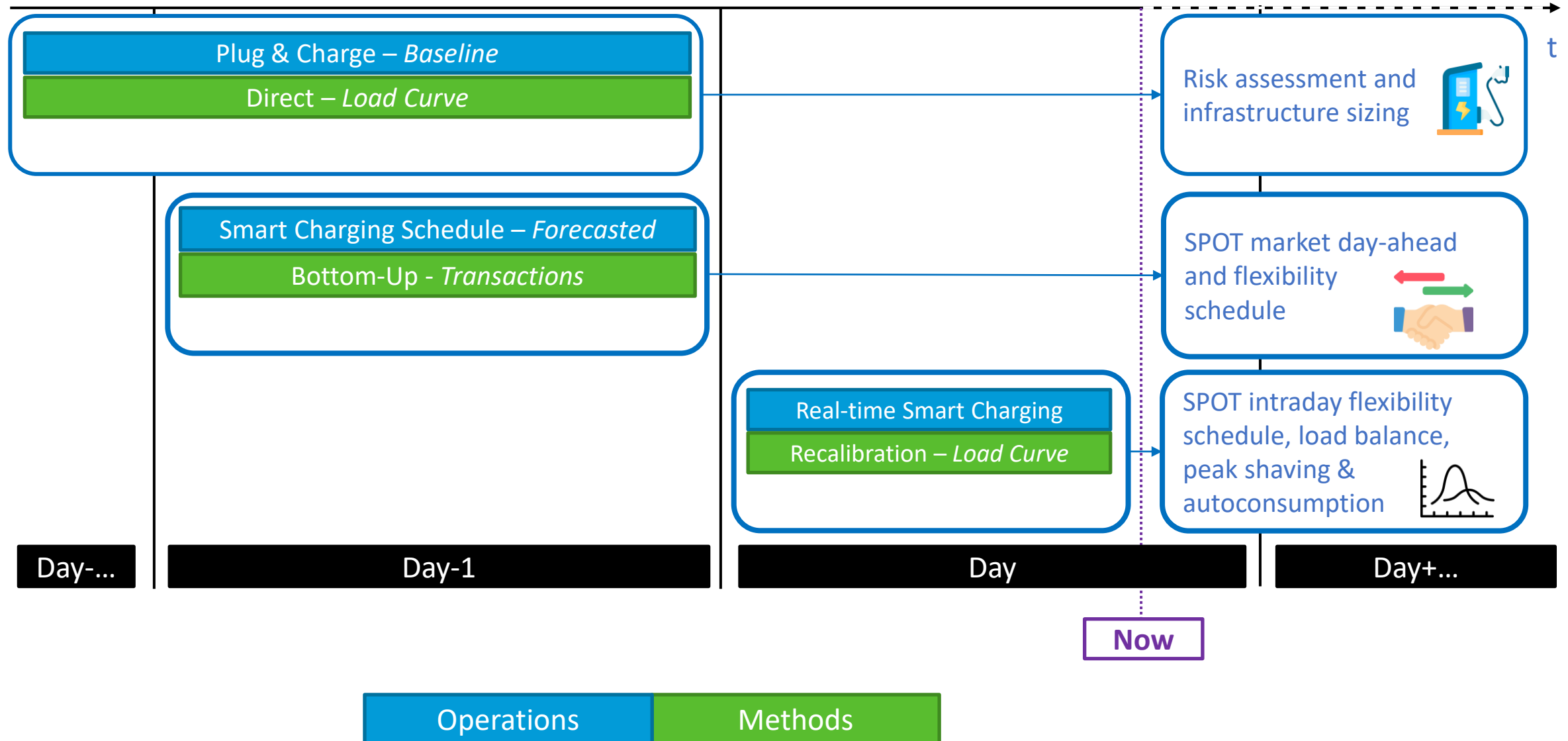


Easy to implement

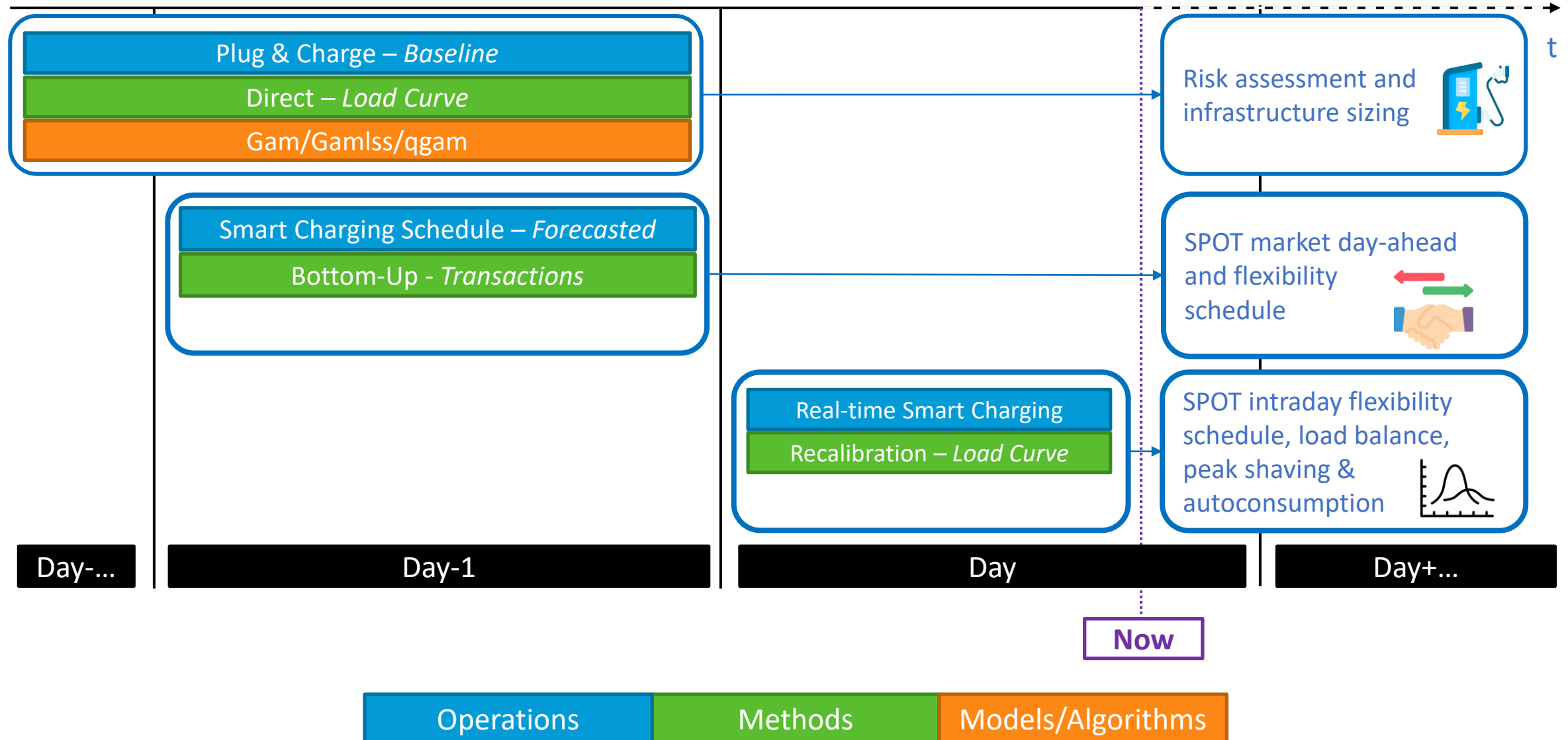




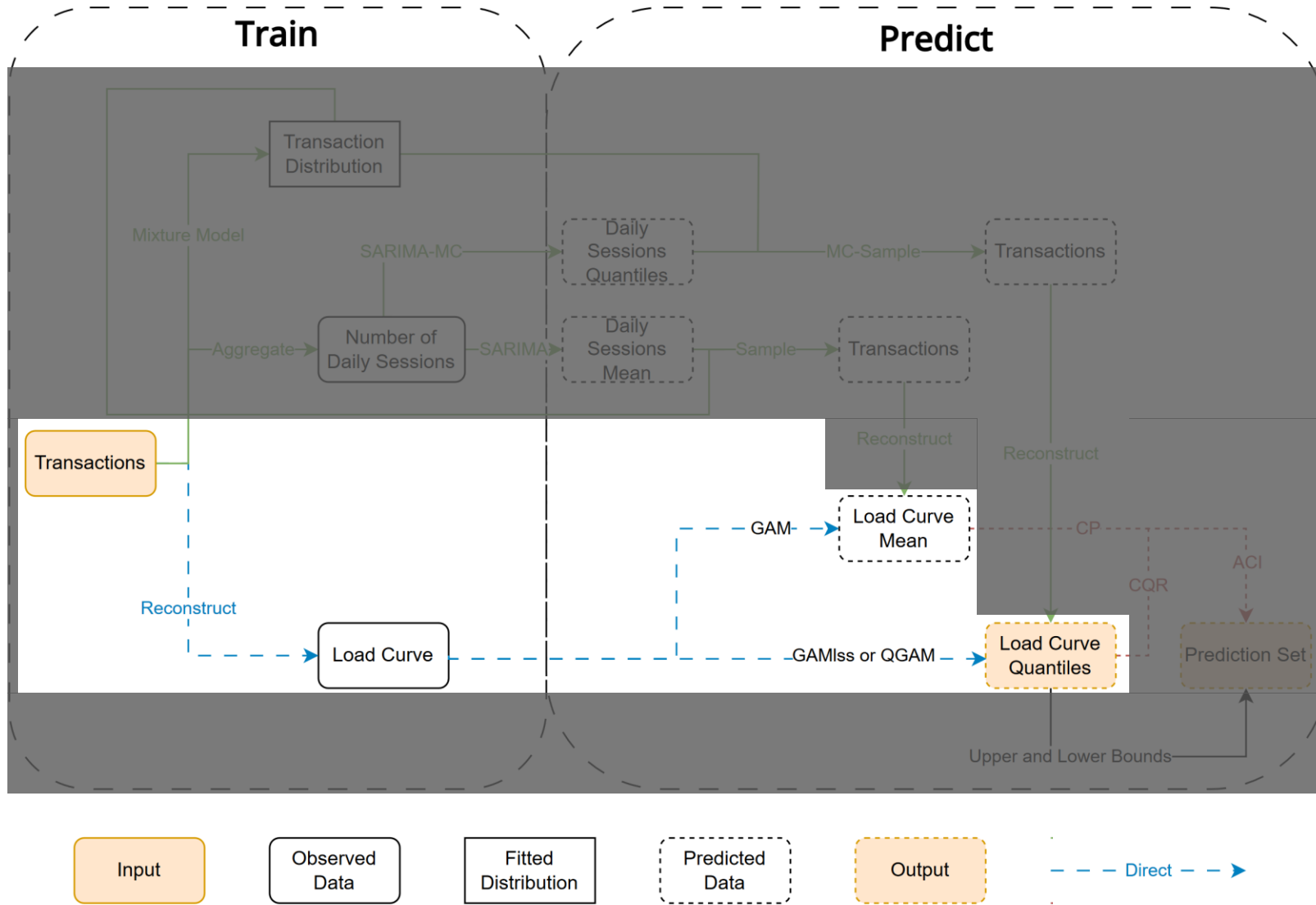
Operations, methods and models across the timeline



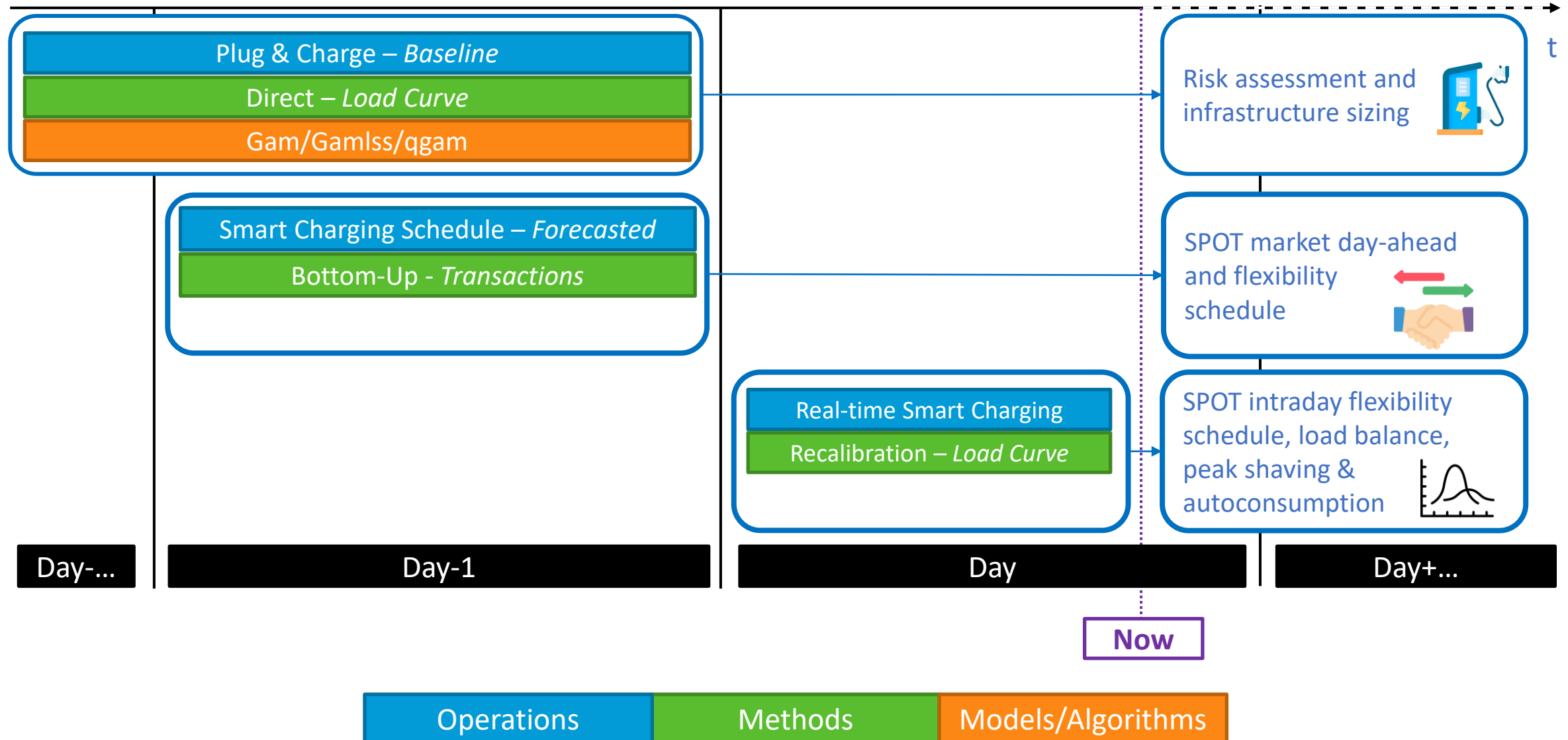
Operations, methods and models across the timeline



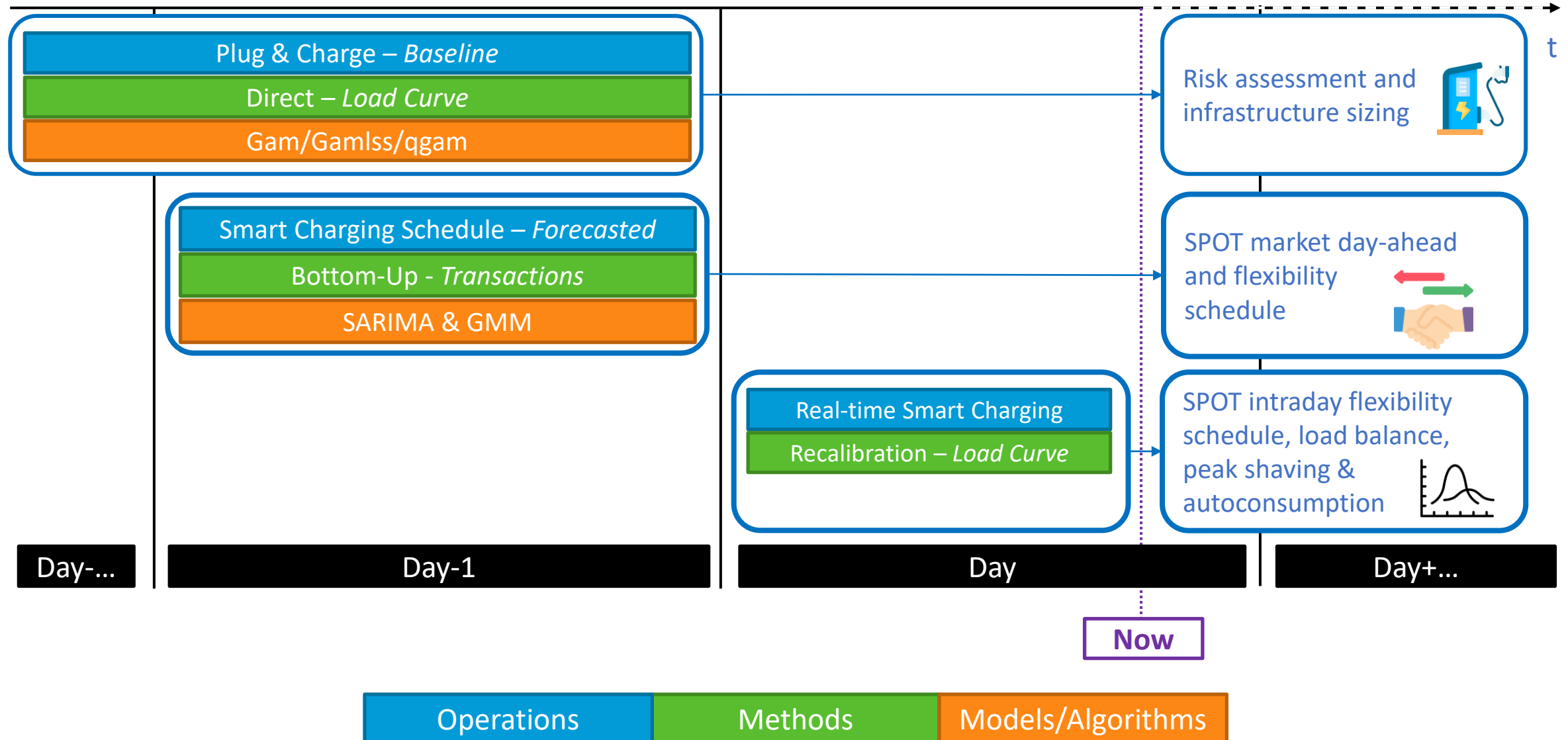
Prediction Framework



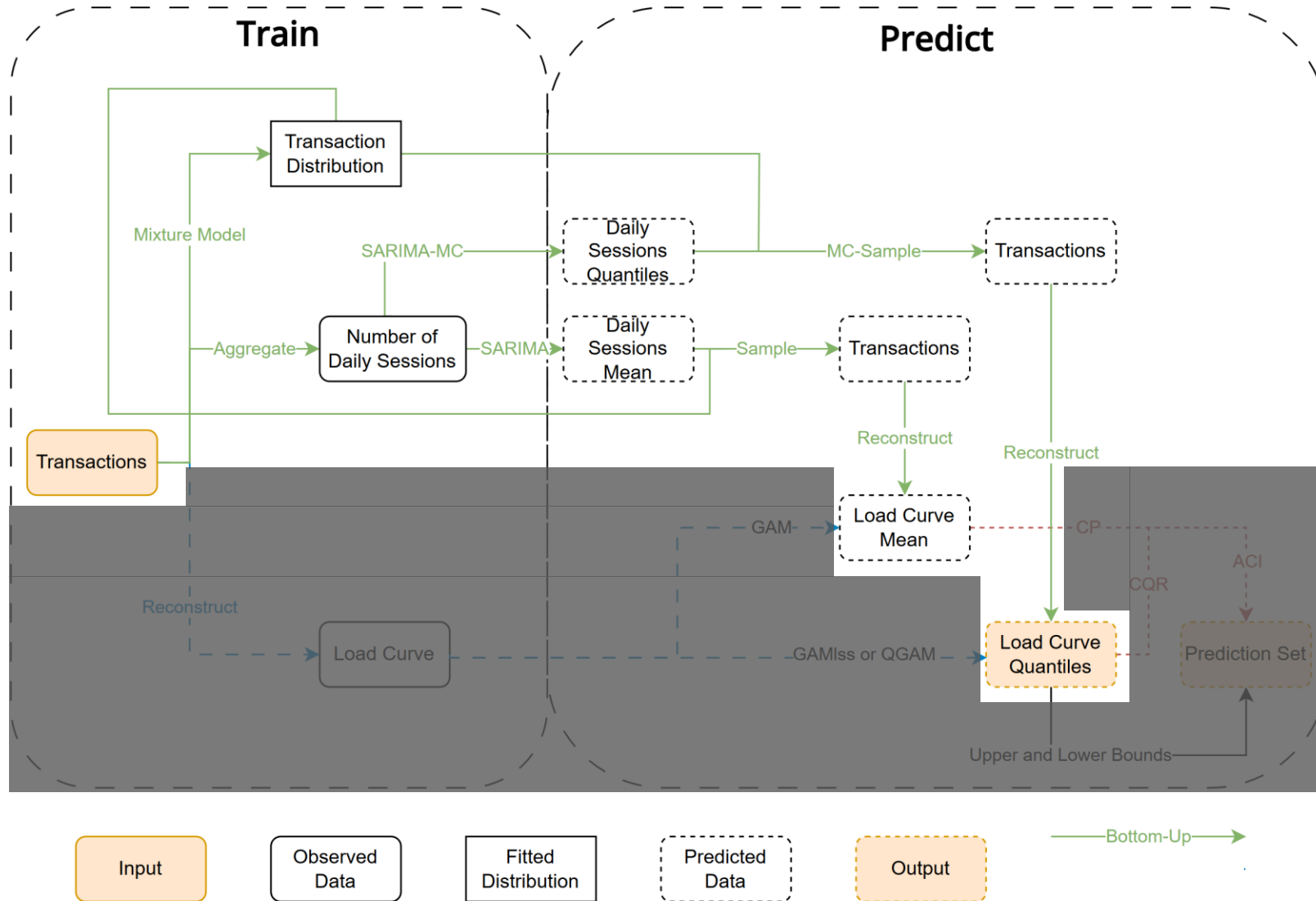
Operations, methods and models across the timeline



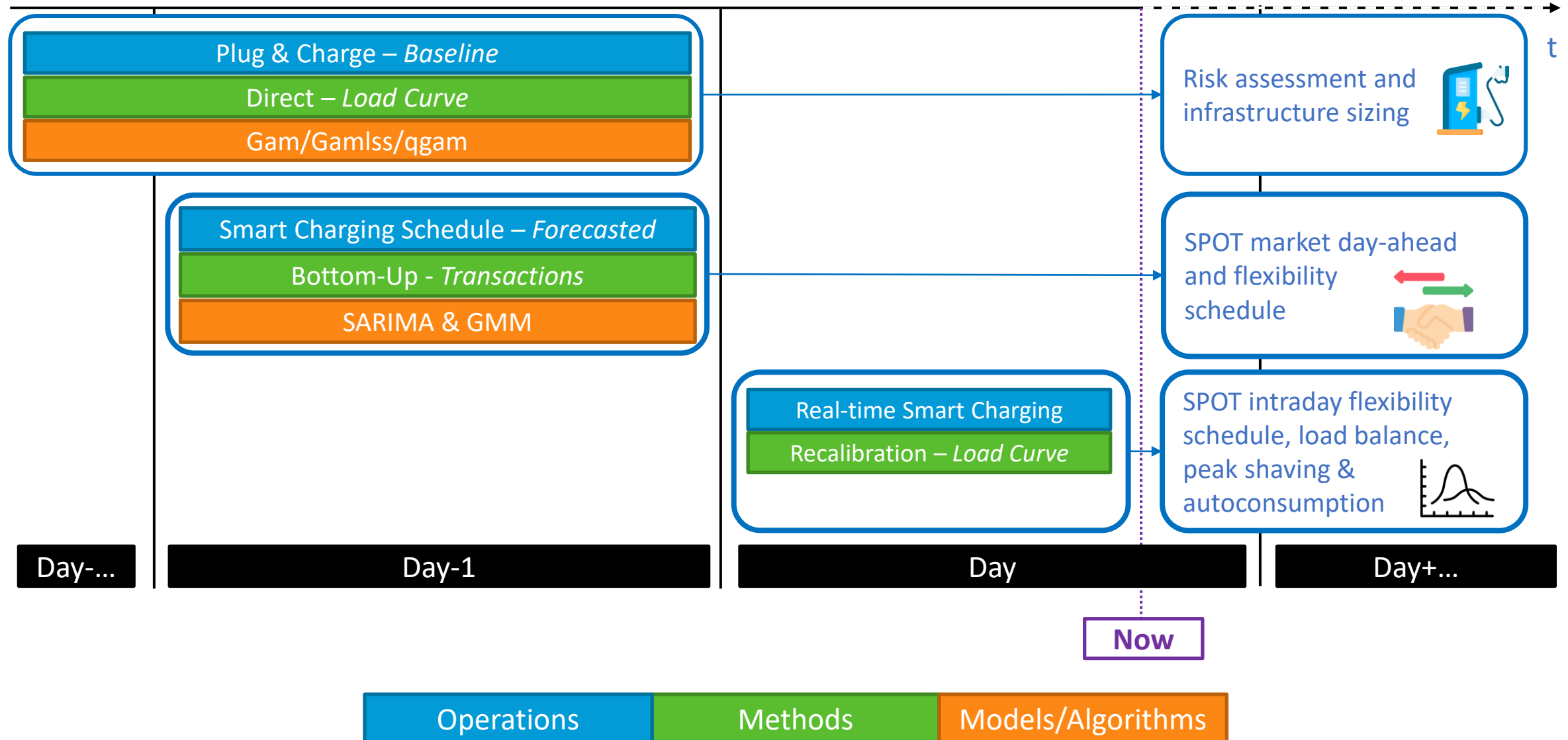
Operations, methods and models across the timeline



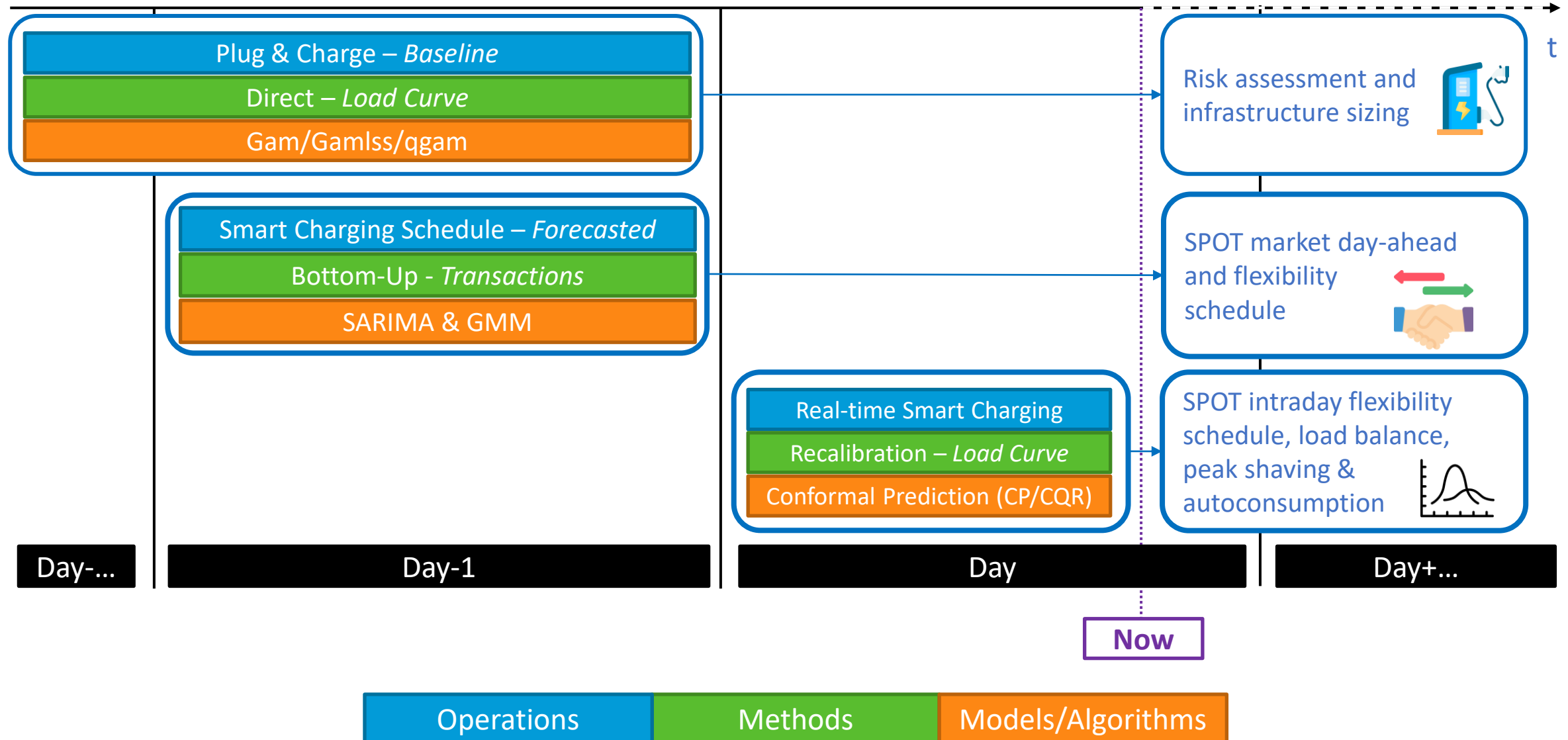
Prediction Framework



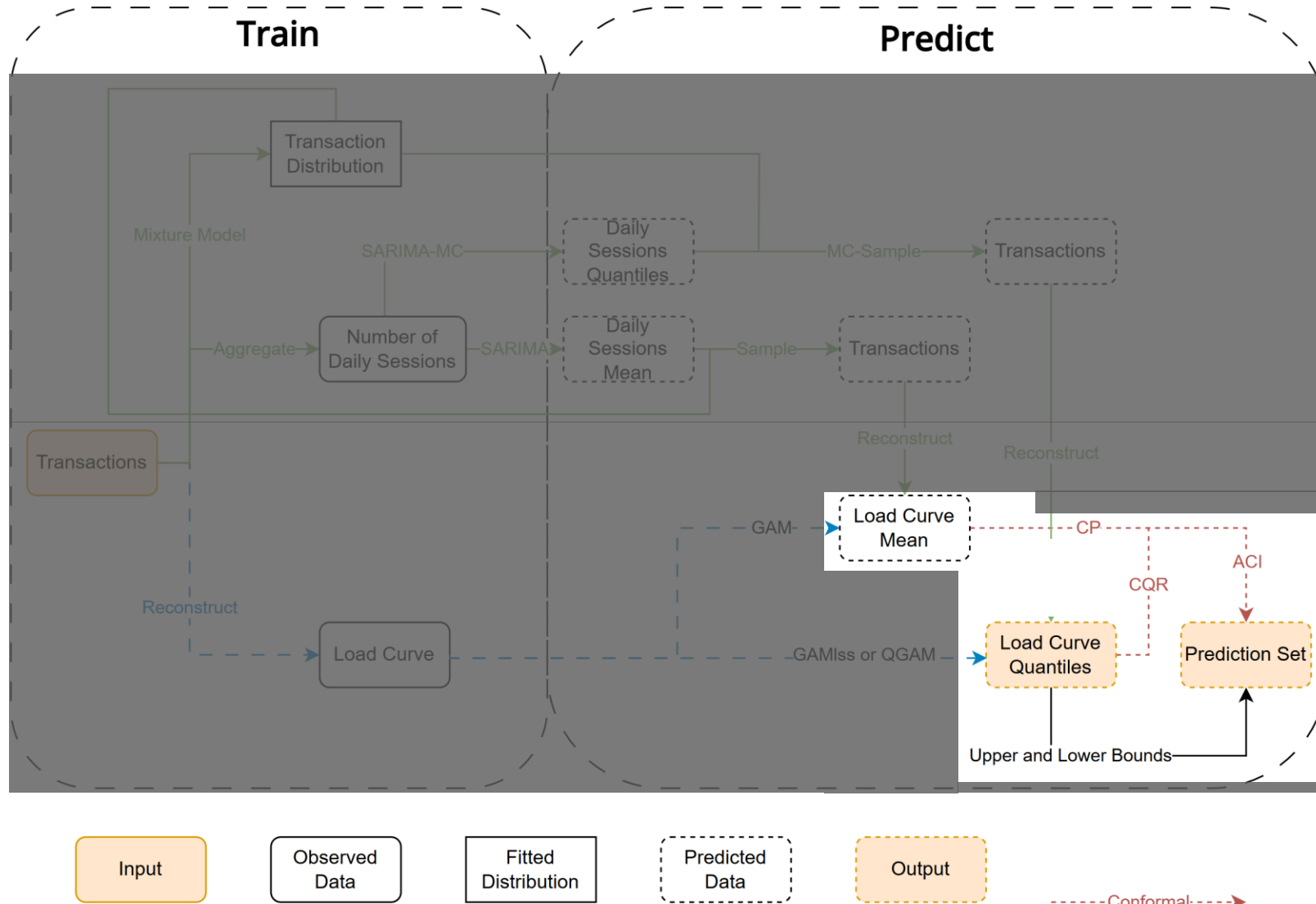
Operations, methods and models across the timeline

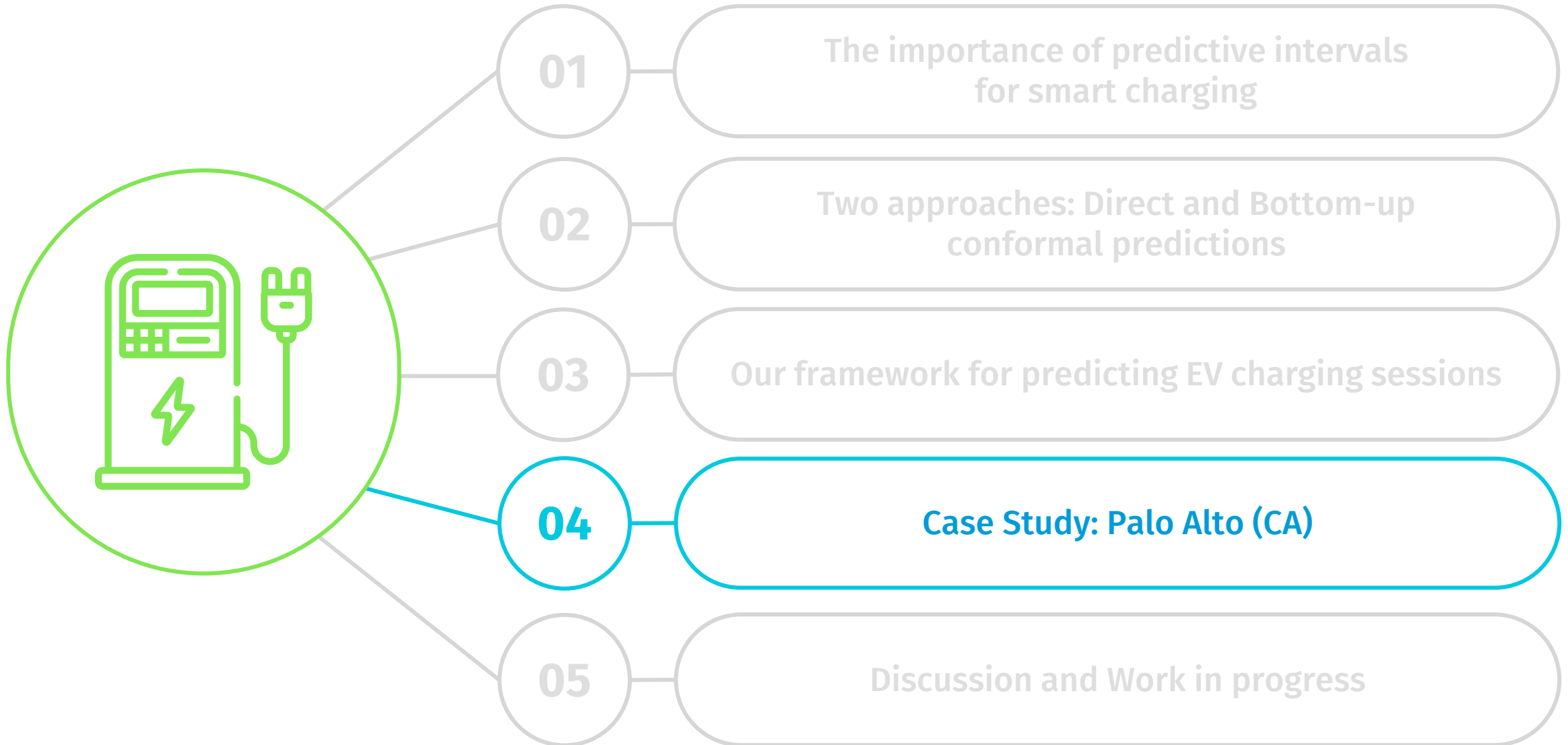


Operations, methods and models across the timeline



Prediction Framework





Palo Alto Dataset

Public Charge

Boulder (USA)

Dundee
(UK)

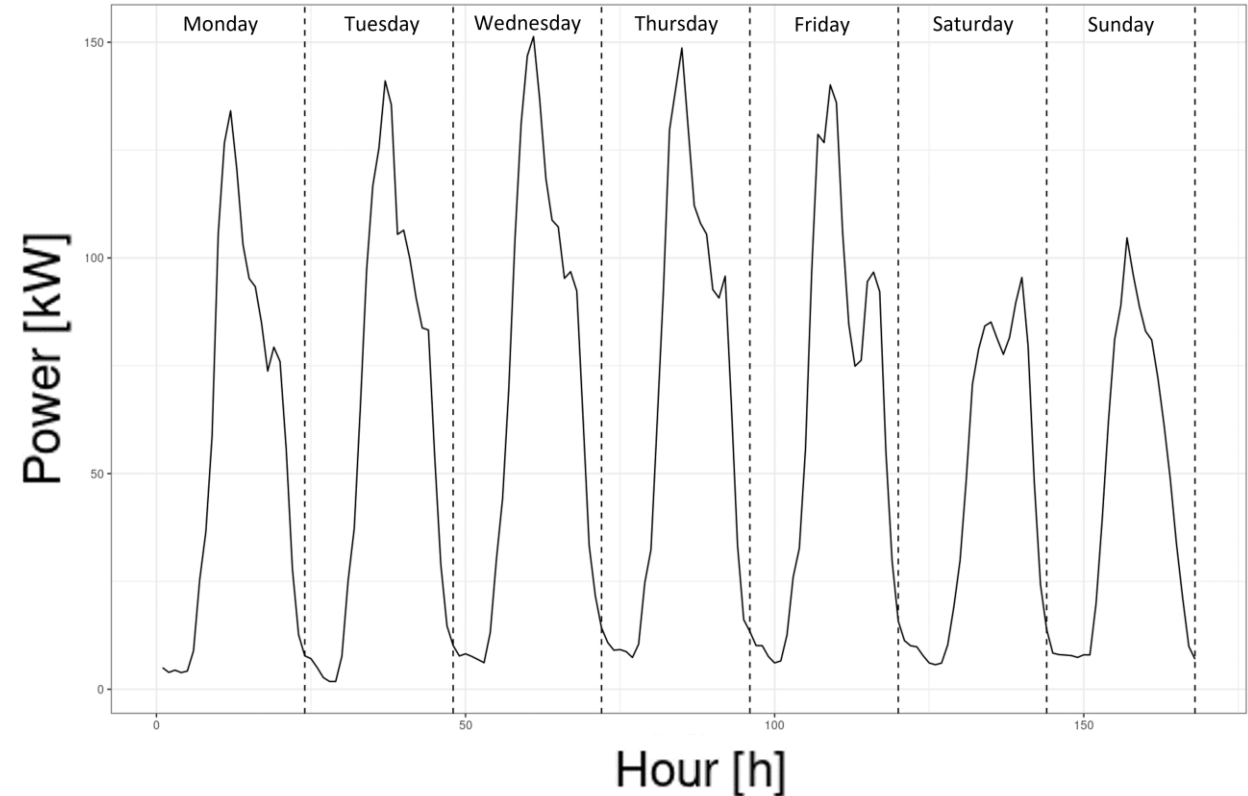
Palo Alto (USA)

Paris
(FRA)

Perth
(UK)



Open Data on public
charge available on GitHub:

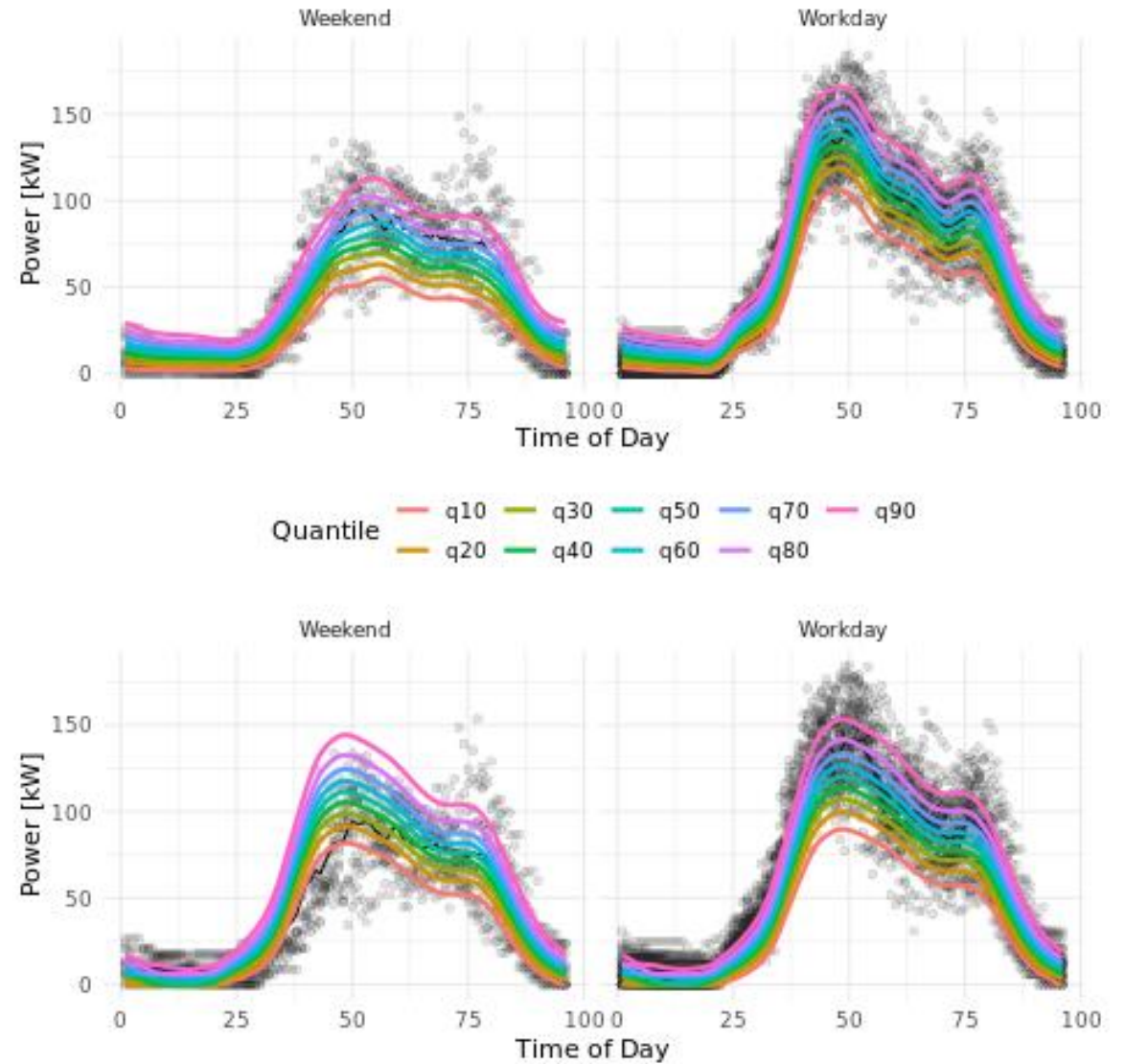


Load profiles for a typical week in Palo Alto (CA)

Uncertainty quantification from Direct and Bottom-Up

Results

- Direct Approaches give the best results in terms of day ahead forecasts for the plug & charge use case
- Bottom-Up gives good results on this task even though it is a more challenging problem
- Among the probabilistic methods, qgam is the best one over almost every quantile, both in accuracy and in sharpness



Averaged quantile forecasts (distinguished by type of day)

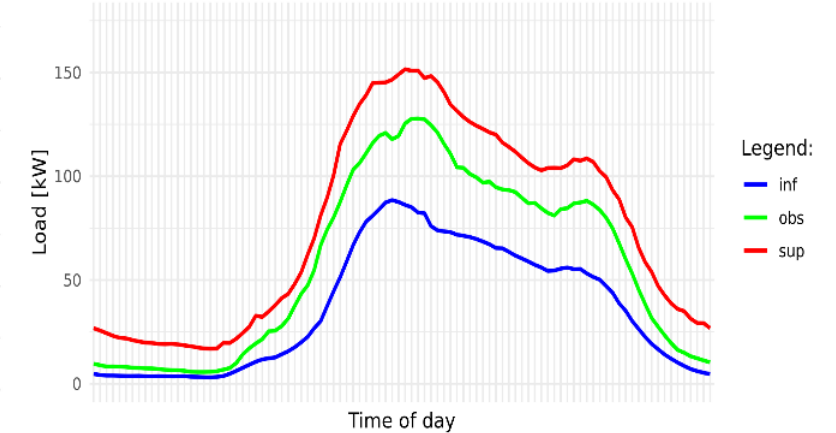
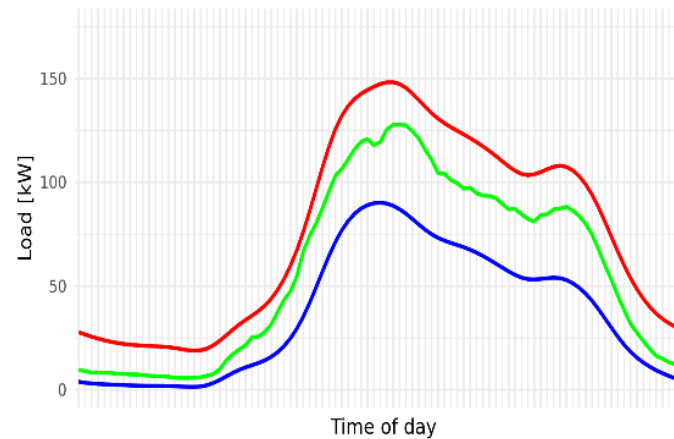
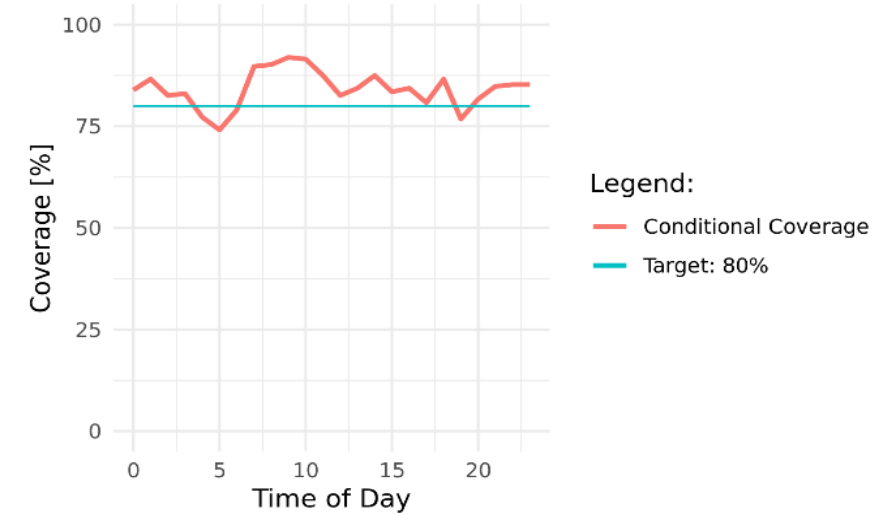
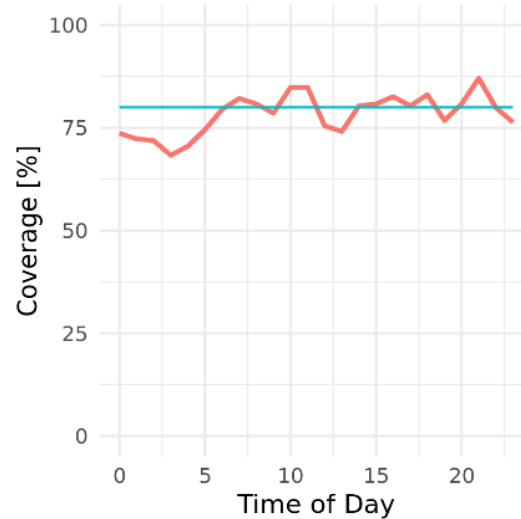
Improvement with Conformal Prediction

From static to adaptive forecasts

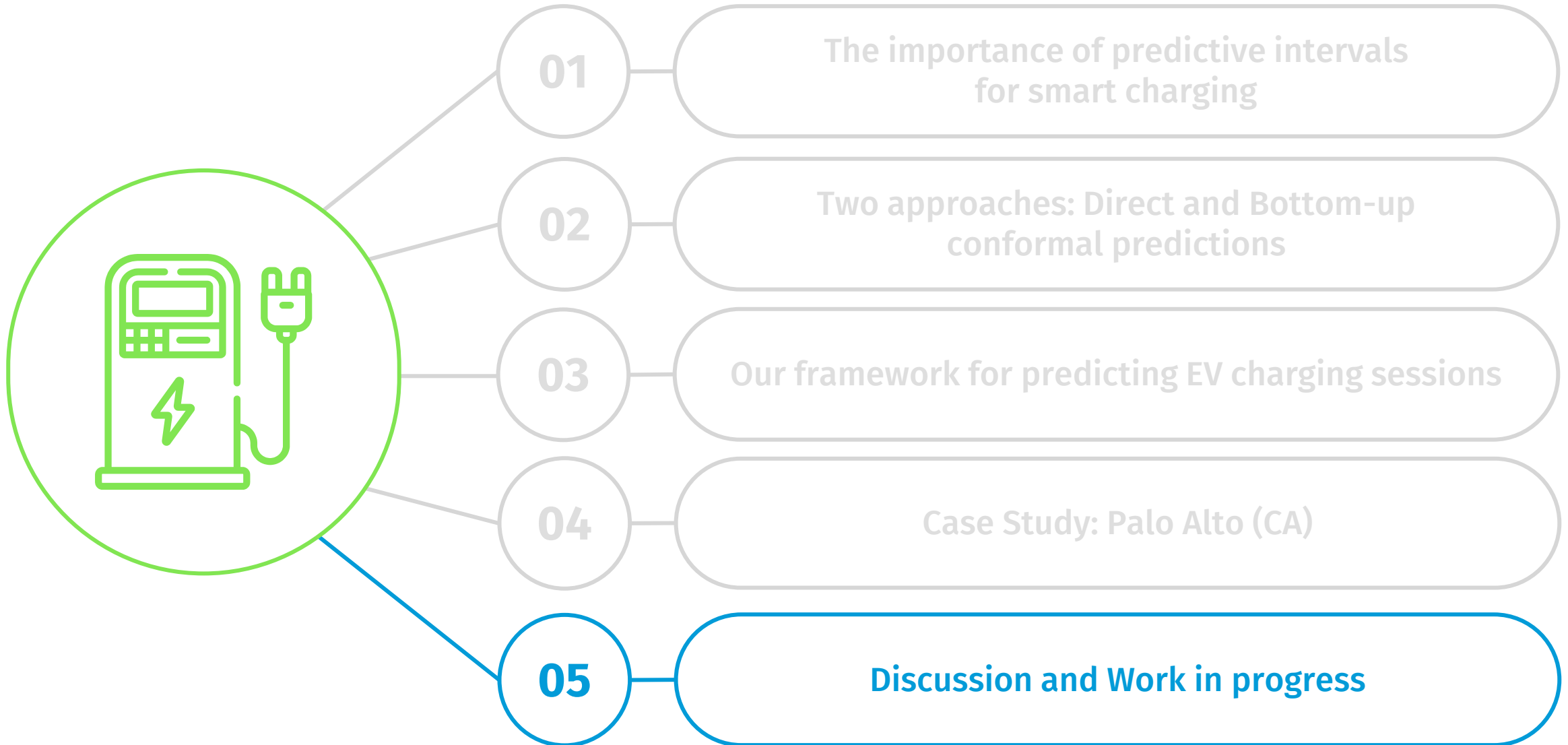
- Both Direct and Bottom-Up methods cannot adapt to behaviour changes
- This can lead to miscalibrated uncertainty in the real-time setting
- To adress this issue, we use Adaptive Conformal Inference methods

Results

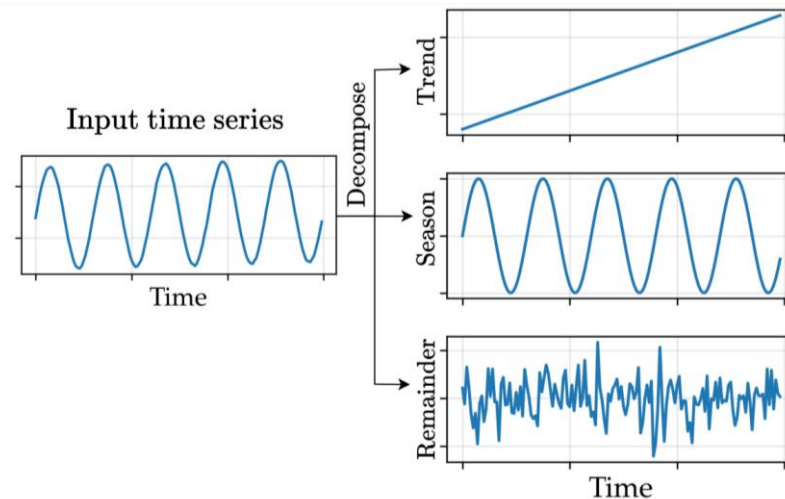
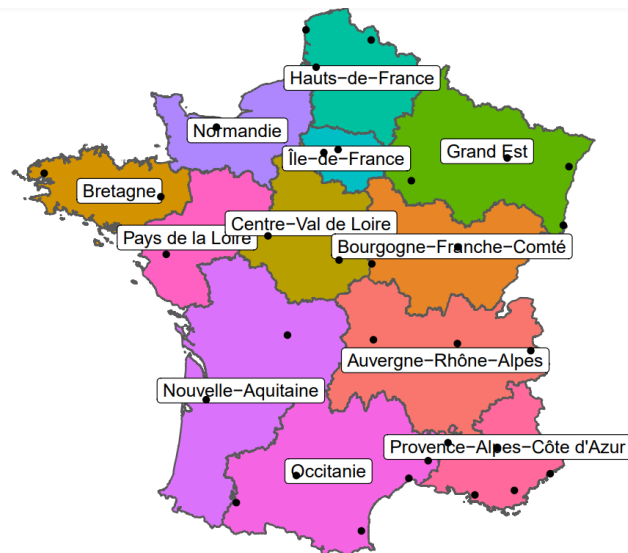
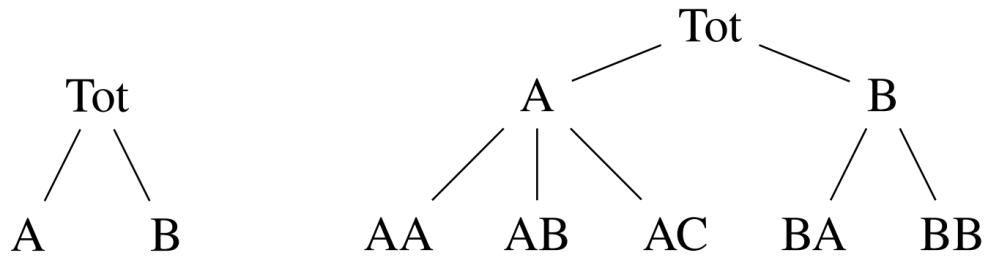
- Higher Empirical Coverage while maintaining conditional Coverage
- Equivalent interval width on daily profile, even sharper during night



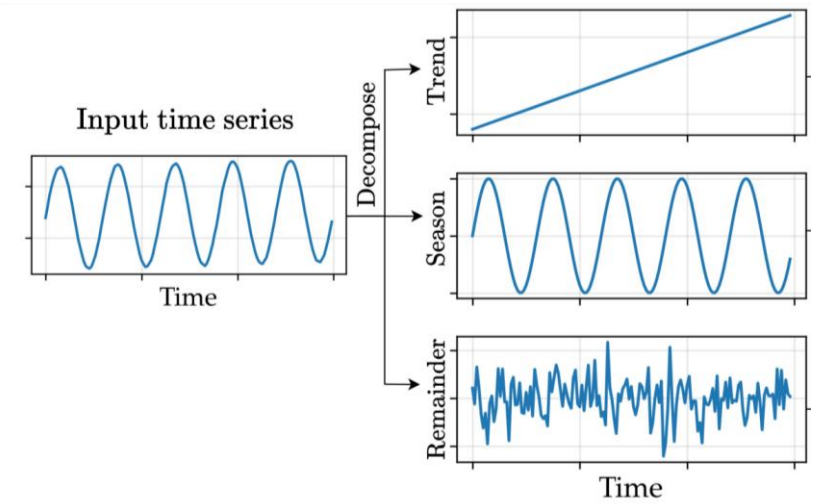
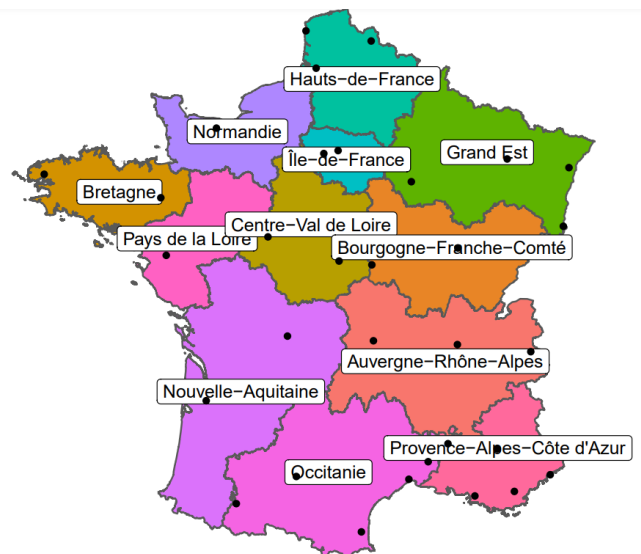
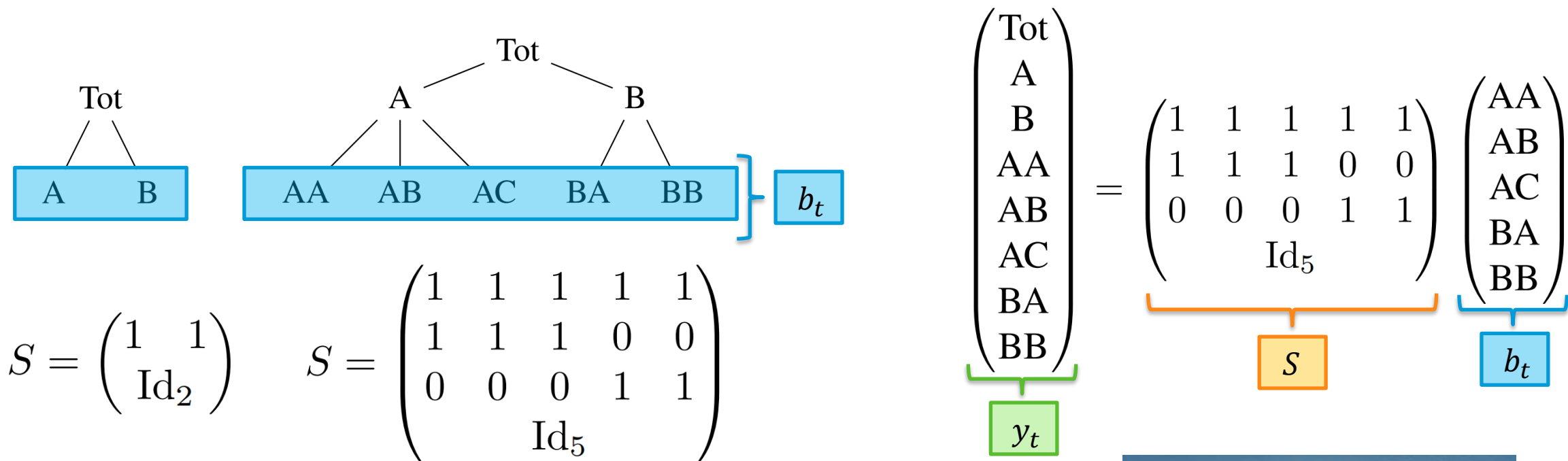
Conditional Coverage and daily profiles for qgam and CQR qgam



Perspectives: How to leverage the hierarchical structure?

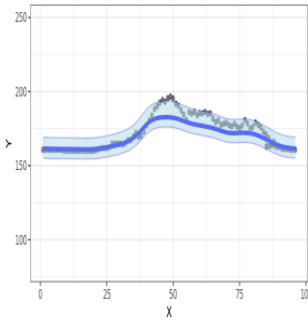
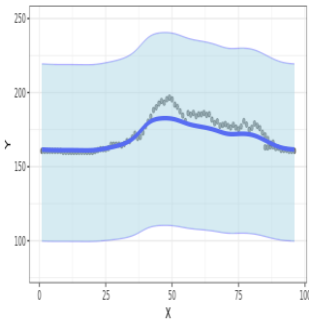
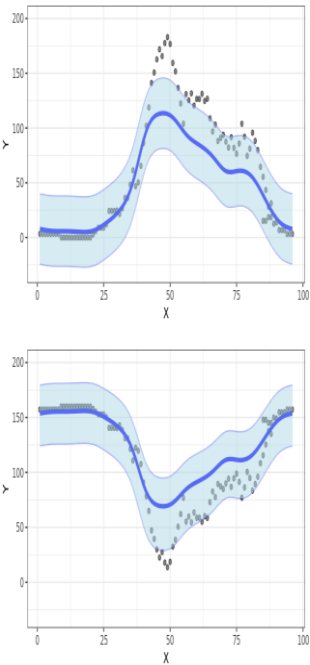


Perspectives: How to leverage the hierarchical structure?



Perspectives: How to leverage the hierarchical structure?

- A well studied problem in point forecasting (⁵Athanasopoulos et al. (2024))
- Recent interest in probabilistic prediction (⁶Panagiotelis et al.(2023),
⁷Wickramasuriya (2024))



Conformal Prediction links point forecasting and prediction set.

What's it for ? CP already provides valid prediction set. Yet, there is still room for improving their efficiency

⁵Forecast reconciliation: A review/ ⁶Probabilistic Forecast Reconciliation: Properties, Evaluation and Score Optimisation/

⁷Probabilistic Forecast Reconciliation under the Gaussian Framework

Perspectives

